

R&B;: integrated mechanism benchmark

213 training configurations | 231 evaluation views | 27 new trainings completed | 6 September 2026

Complete: 186 prior second-stage results +18 QR control trainings +9 joint-width trainings. The latter yield 27 views of 9 shared checkpoints.

Study A: QR Radon exceeds self-only by 2.63 pp. Its mean advantage over spatial scrambling is only 0.18 pp; the interval does not resolve the difference.

Study B: all 9 joint-width trainings select epoch 0 after reaching the prespecified plateau. Flat width curves here mean no recovered improvement, not successful robust communication.

Working conclusion: independently trained noncentered-SVD Radon remains the most balanced configuration in this benchmark. Correct spatial structure has stronger support under SVD than QR.

Scope: historical development data; 296 participants, not 231 independent datasets. This is controlled mechanism evidence, not external clinical validation.

How to read this integrated report

The preserved 186-result study and two follow-ups retain separate comparison families.

Pages 4-54 preserve the 186-result stage report: common methods, geometry, direction, pairing, diagnostics, 31 primary contrasts and the complete 186-row table.

Pages 55 onward contain the new QR mechanism and joint-width studies, including every new training/view result. All 231 evaluation views are also supplied as CSV.

Study A and Study B each use their own three-contrast max- $|t|$ family. The original 31-comparison family is unchanged. 10,000 draws share participant indices.

The ± 1 pp research band is a prespecified interpretation convention. An interval excluding zero need not clear the +1 pp substantive-improvement boundary.

Original 23-page report and the short project-introduction deck remain separate. No historical result, failed attempt or completed study is overwritten.

What the completed follow-ups establish

All effects below average model F1 equally over the same three seeds, three widths and two branches.

Contrast	Effect (pp)	Simultaneous95% interval	Interpretation
QR Radon - self	+2.63	[+0.24, +5.02]	unresolved
QR Radon - scrambled	+0.18	[-1.12, +1.48]	unresolved
SVD - QR spatial gain	+1.79	[+0.03, +3.56]	unresolved
Joint - independent: learned channel	-2.90	[-5.33, -0.46]	unresolved
Joint - independent: SVD	-4.19	[-7.17, -1.22]	supports substantive decline
Joint - independent: QR	-2.63	[-4.85, -0.41]	unresolved

Structured cross-source communication

Souray Meng | R&B; (Radon Bridge) | Three-seed, three-width controlled benchmark

186 accepted second-stage results: 129 historical references +57 prespecified supplemental trainings. Existing evidence remains intact.

Four questions: additional value of Radon geometry; dependence on the fixed subspace; direction and correct pairing; comparison with common communication methods.

Research boundary: same glaucoma endpoint, paired CFP/OCT, existing task-adapted native networks. No extra pretraining or basis fitting in this supplement.

Interpretation: controlled mechanism evidence on a historically selected development set. This is one component of a future paper, not complete NBE submission evidence.

Networks, communication and shared training

Original MHD V4 stages, task heads, CE losses and Node IDs are preserved.

CFP input [B,2,3,224,224]; OCT input [B,2,1,32,96,96]. Both complete ResNet18 task networks were separately trained to plateau.

Restore the same per-seed CFP/OCT best checkpoints, insert one stage3 communication adapter, then train all native and adapter parameters. BatchNorm continues updating.

Native learning rate $6e-5$; heads/communication $1e-4$. Batch16, AdamW, weight decay0.01; separate gradient clipping at5.

At least8 epochs; plateau after6 epochs without improvement greater than0.001. LR x0.3 after3 stagnant epochs. Sixty epochs is a guard, not convergence.

A common second-stage checkpoint is selected by the mean of the two branch macro-F1 scores. No test data are read.

Prespecified experiment matrix

Seeds3416/3417/3418; rho is a dimension ratio. Common baselines do not have a rho axis.

Supplement	Configurations	Training tasks
Random QR / linear resampling	3 seeds x3 widths	9
SVD and QR / directional Radon	2 bases x2 directions x3 seeds x3 widths	36
MMTM single-stage adaptation	Reduction4/8; hidden256/128; 3 seeds	6
Bidirectional cross-attention	Dimension128/256; 4 heads; 3 seeds	6
Reused historical results	Same parents, bases, predictions and converged summaries; hashes verified	129

What is controlled, and what is not

Matching training conditions does not imply identical capacity or computational cost.

Radon paths: stage3, M32, S64; rho1/16,1/8,1/4 gives channel rank16/32/64 and per-source mixer width512/1024/2048.

Ordinary communication uses fixed linear resampling with matched forward/return operator row L2 norms. Packing factor32 is not an angle count; rank and singular spectrum are not matched.

Directional training retains both self blocks and masks one cross block. Stored and effective parameters are reported separately; effective capacity is not fully matched.

MMTM and attention retain their required nonlinearities. They are single-stage communication adaptations, not reproductions of complete published task systems. Both predeclared configurations are reported.

Historical search budgets differ. Existing results are not relabeled as new prespecified independent validation.

Statistical analysis and decision rules

31 fixed primary comparisons; branch, width and seed strata remain visible.

Seven mechanism contrasts average each model F1 equally across three seeds and three widths. Twenty-four common-baseline contrasts average across seeds for each fixed basis/width/configuration.

10,000 paired bootstrap draws resample the same 296 participants for all models. Calculate model F1 first, then average; do not ensemble across seeds or widths.

Ordinary 95% percentile intervals and approximate 31-contrast max- $|t|$ simultaneous intervals are both reported. Zero bootstrap variance is marked undefined.

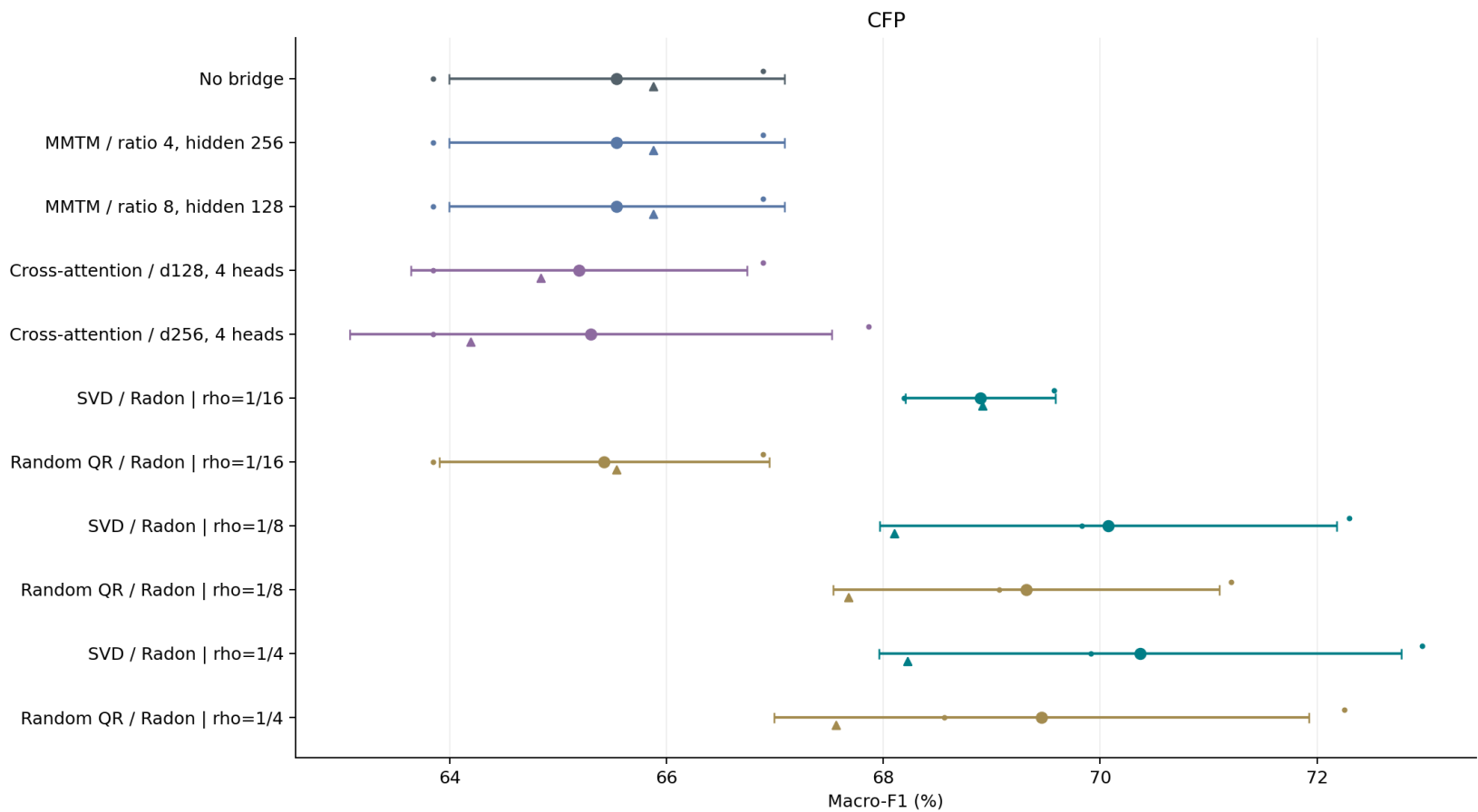
Research reference band ± 1 percentage point: the simultaneous interval supports substantive improvement, substantive decline, practical similarity, or remains unresolved.

Intervals condition on fitted models and selected checkpoints. Three seeds and 20 derangements do not increase the independent participant count.

Common communication methods: CFP

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Common communication methods: CFP

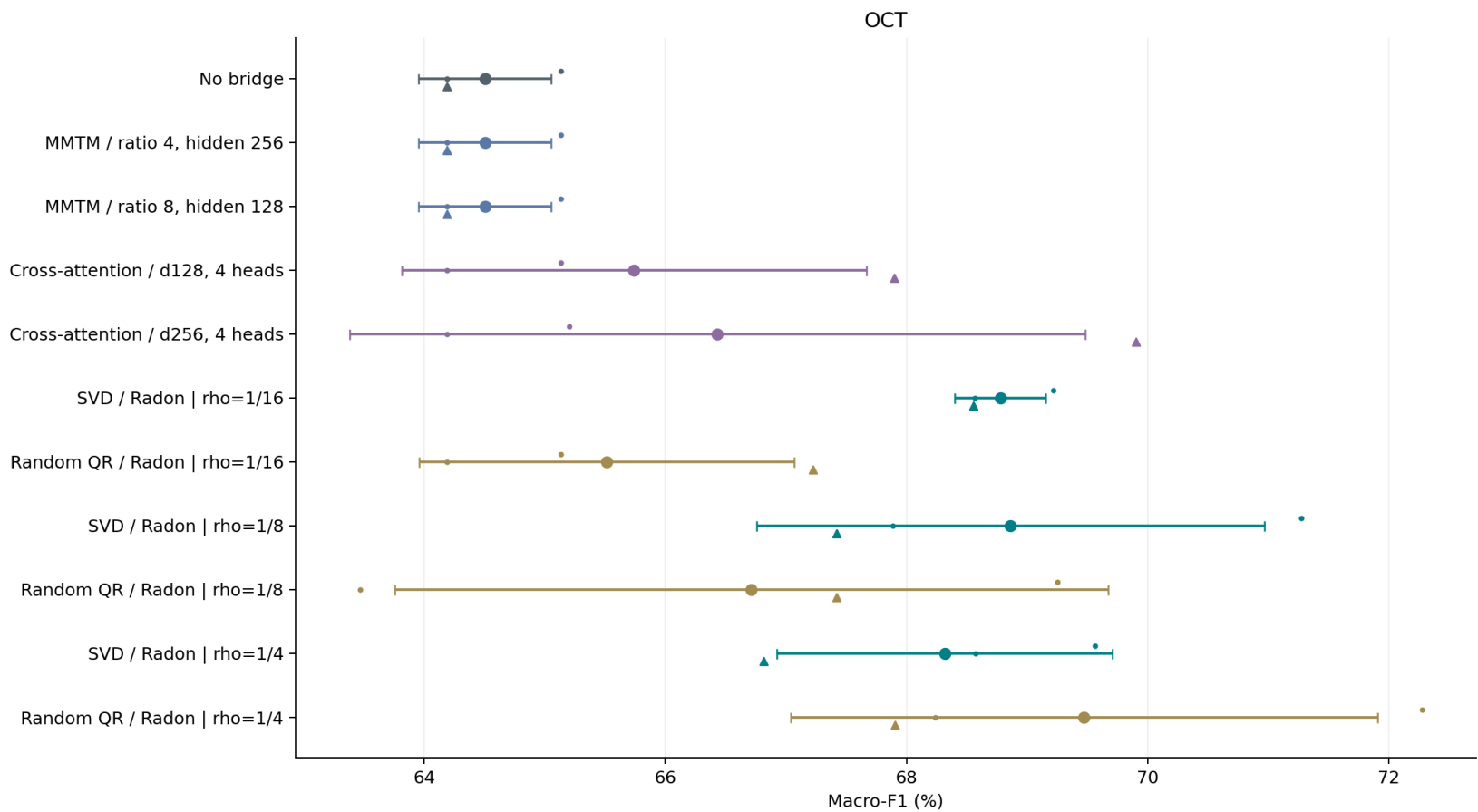


Mean +/- sample SD over three seeds. Points show individual seeds; triangles mark 3418. Parameters and search histories differ.

Common communication methods: OCT

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Common communication methods: OCT

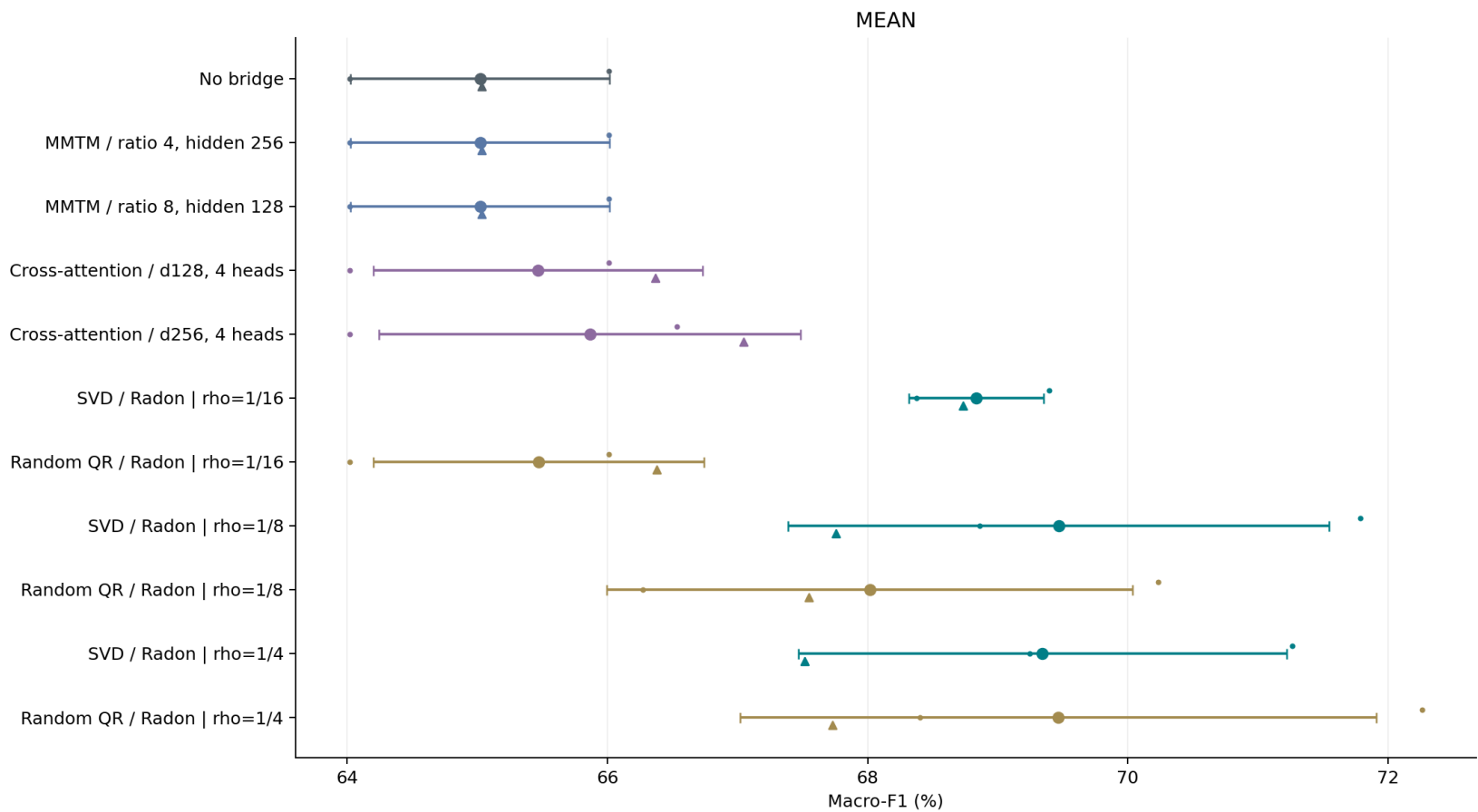


Mean +/- sample SD over three seeds. Points show individual seeds; triangles mark 3418. Parameters and search histories differ.

Common communication methods: MEAN

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R&B (Radon Bridge) | Common communication methods: MEAN

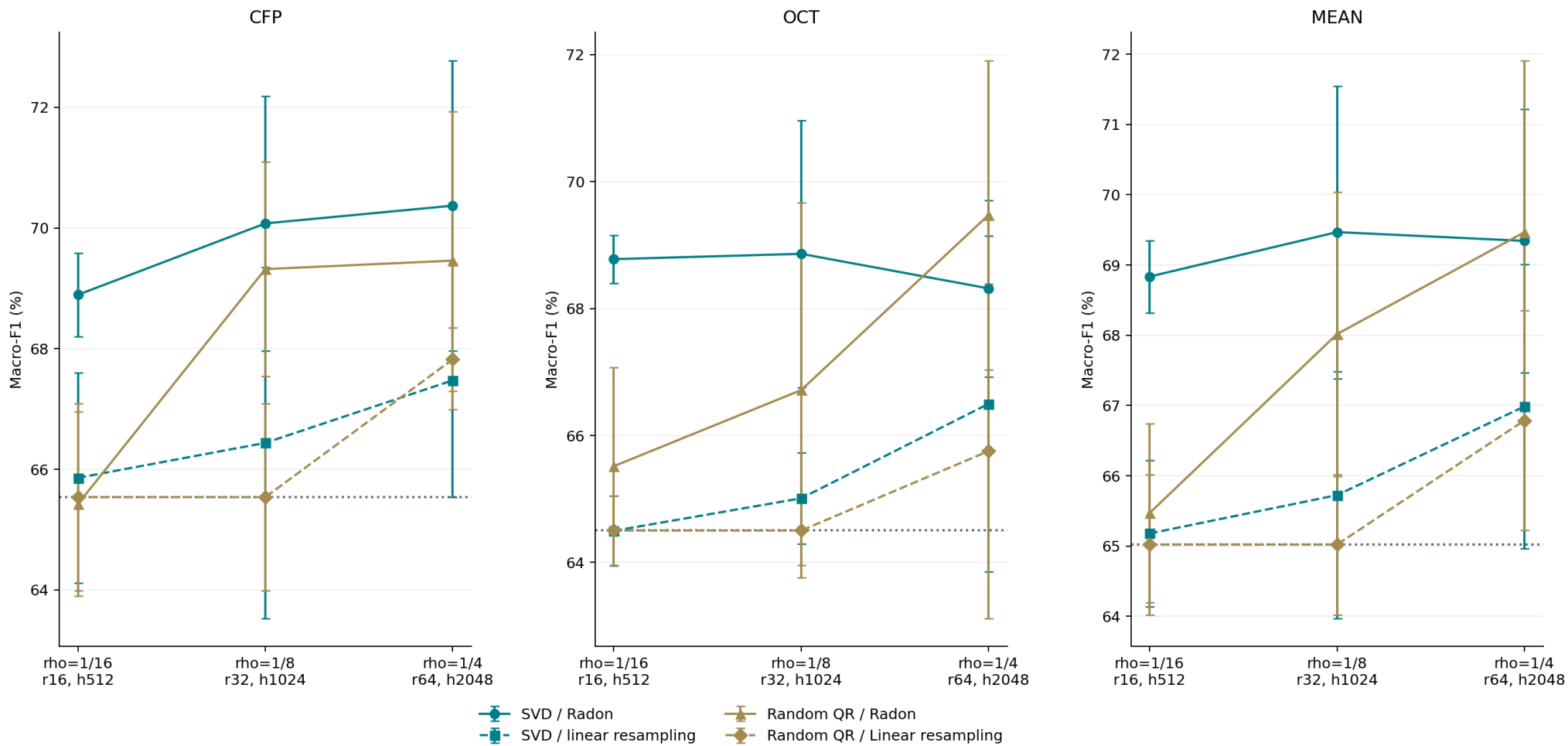


Mean +/- sample SD over three seeds. Points show individual seeds; triangles mark 3418. Parameters and search histories differ.

Fixed basis x communication geometry

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Fixed basis x communication geometry

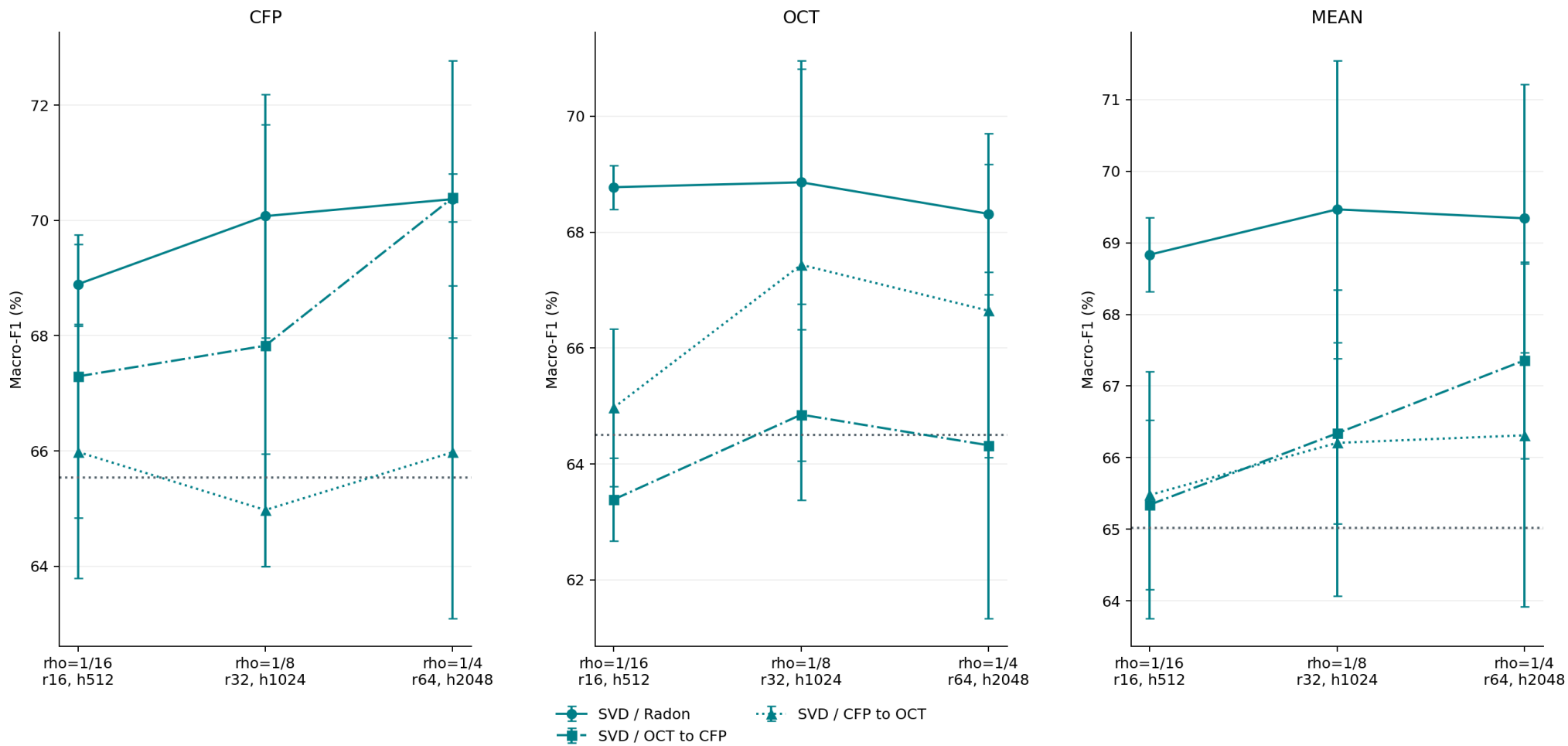


Three-seed mean +/- sample SD. CM rank r is not applicable; its h matches. Resampling factor32 is packaging, not an angle count.

SVD: directional training

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | SVD: directional training

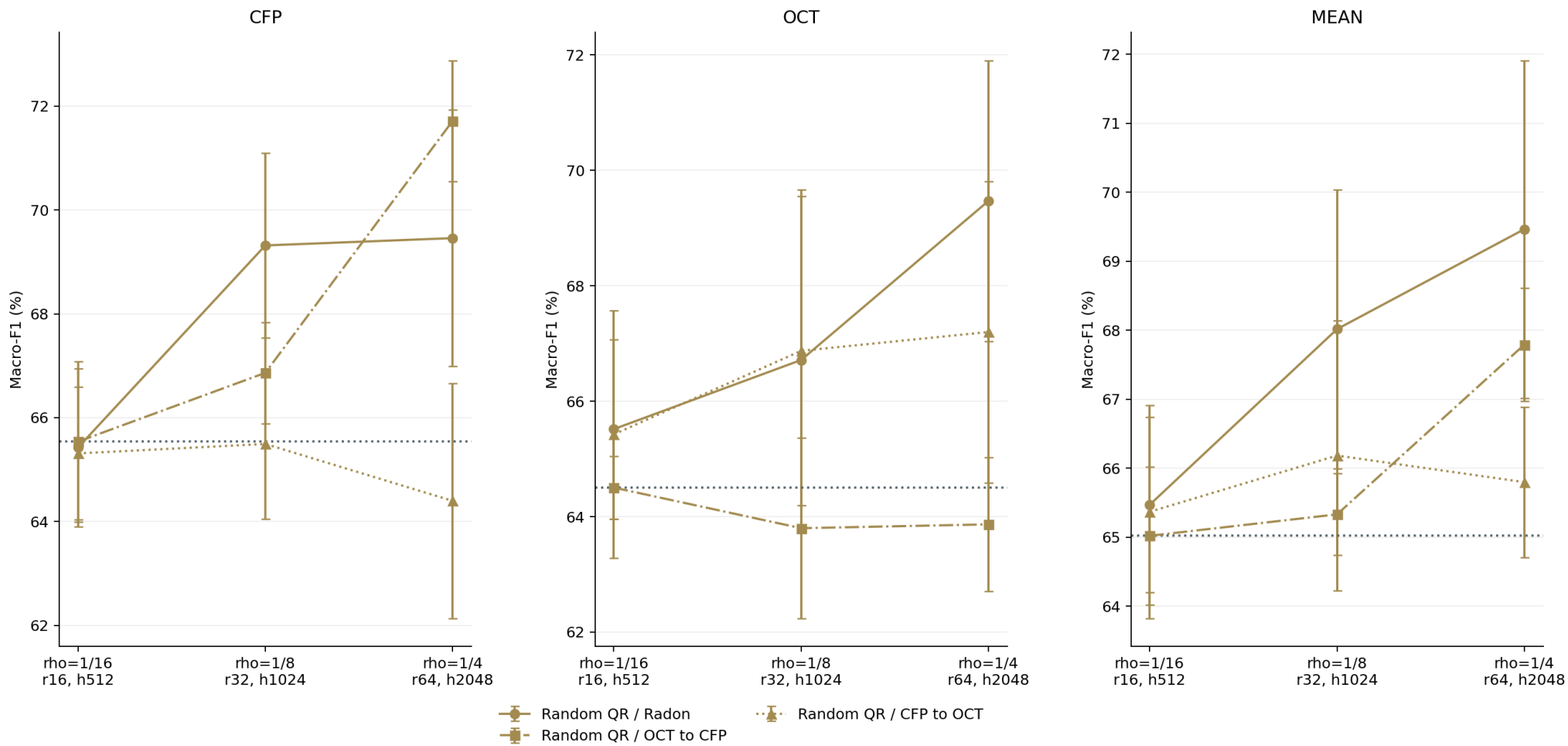


Three-seed mean +/- sample SD. CM rank r is not applicable; its h matches. Resampling factor32 is packaging, not an angle count.

Random QR: directional training

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Random QR: directional training

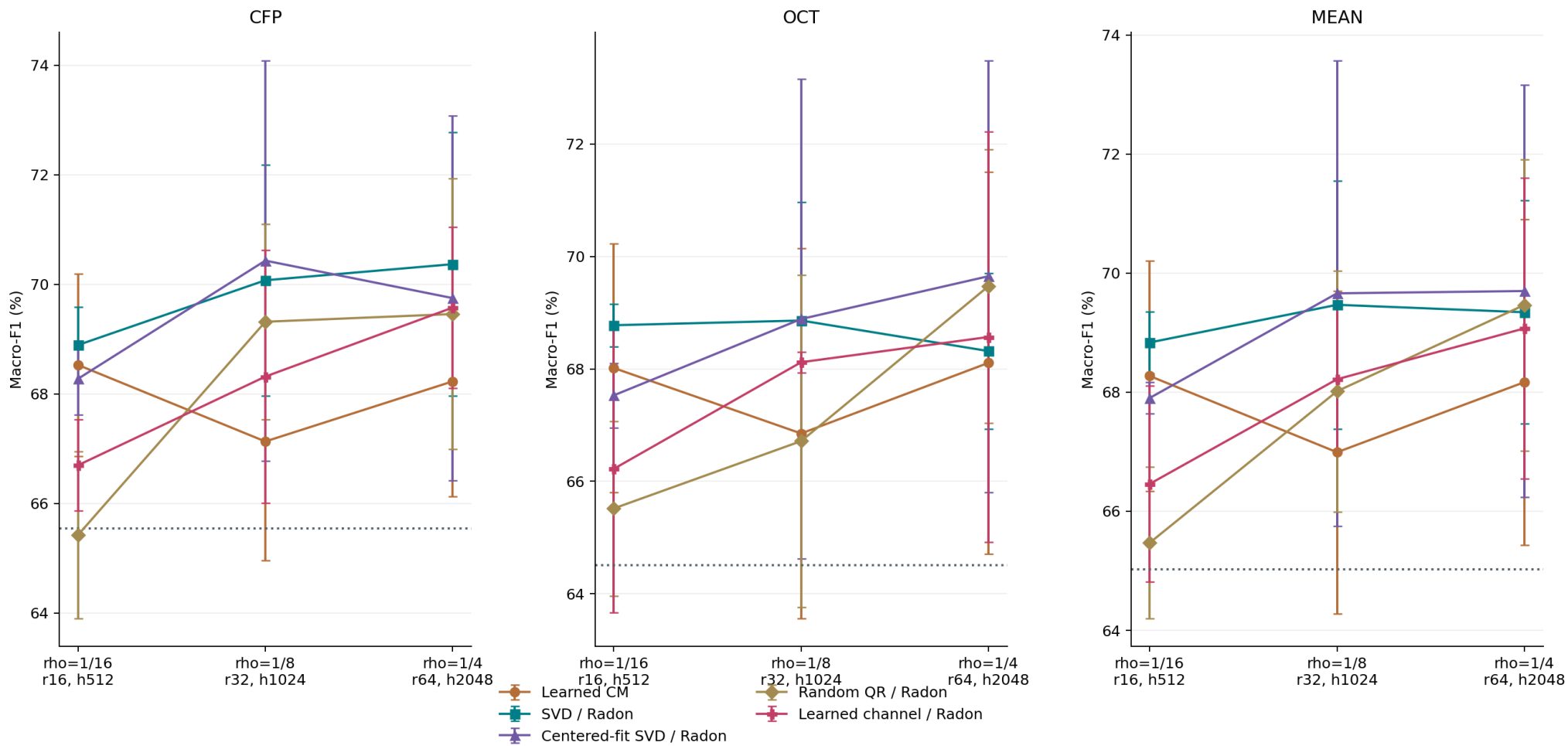


Three-seed mean +/- sample SD. CM rank r is not applicable; its h matches. Resampling factor32 is packaging, not an angle count.

Historical compression exploration

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Historical compression exploration

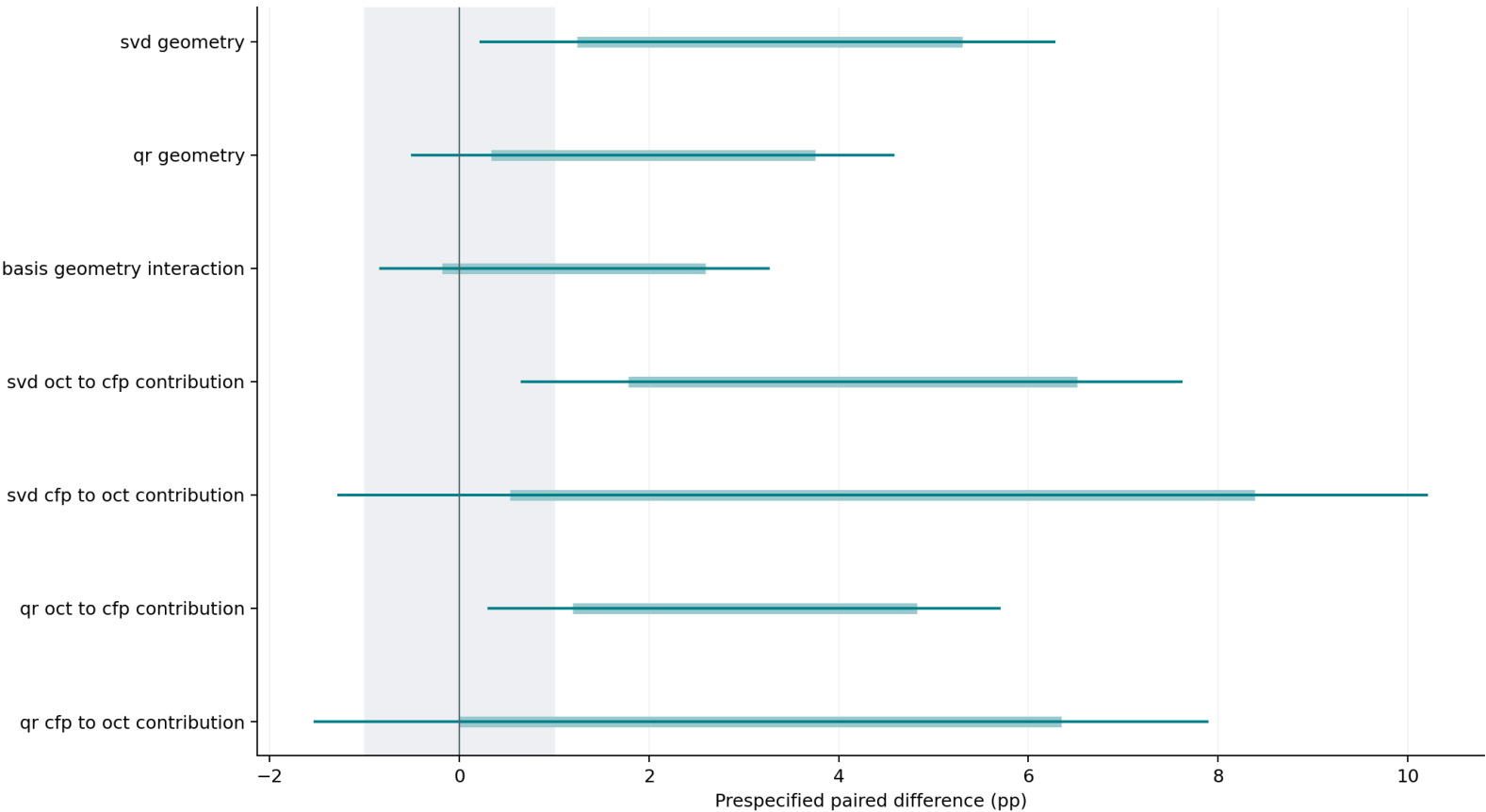


Three-seed mean +/- sample SD. CM rank r is not applicable; its h matches. Resampling factor32 is packaging, not an angle count.

Primary contrasts with simultaneous uncertainty

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Primary contrasts with simultaneous uncertainty

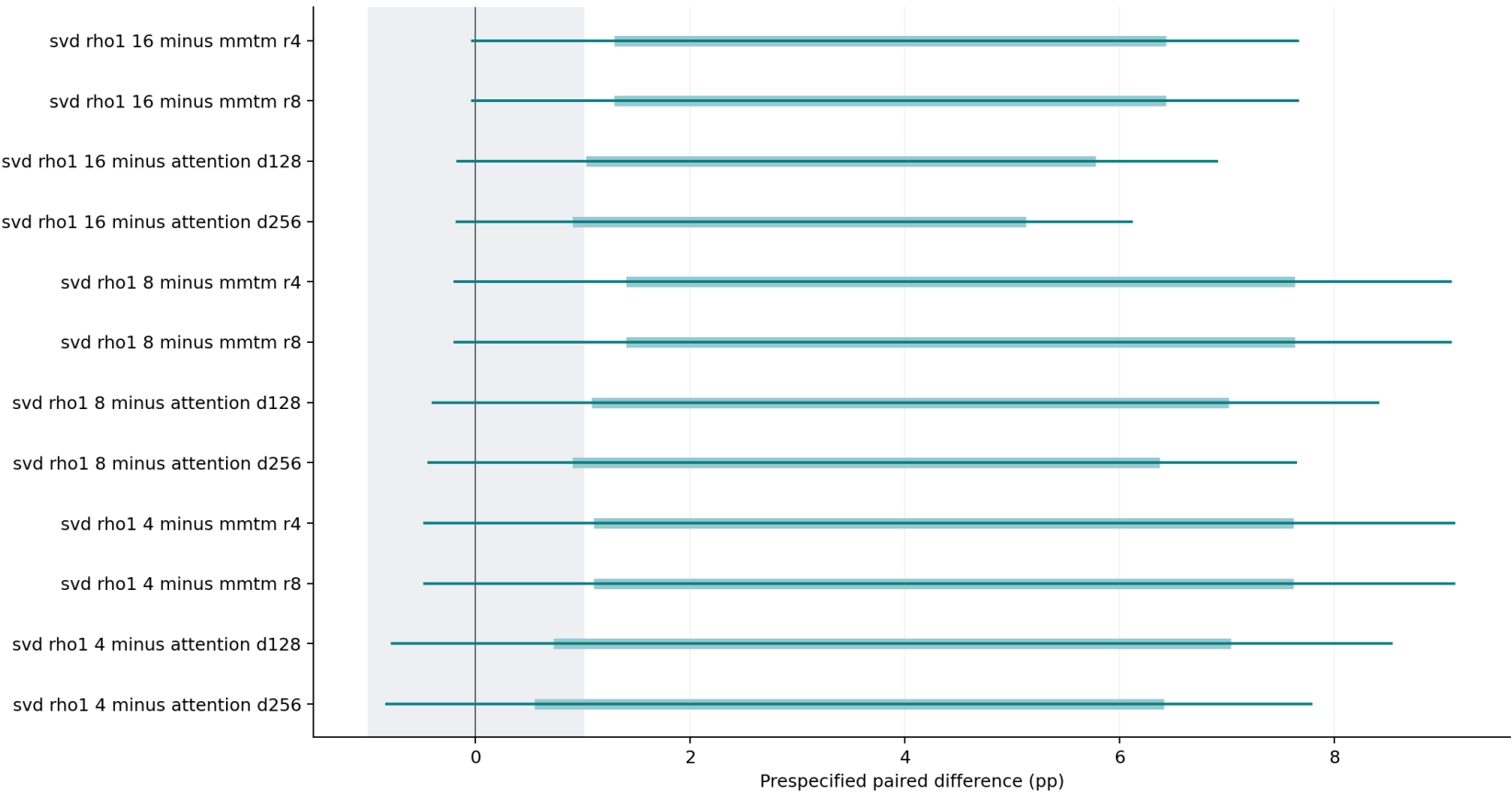


Thin: max-|t| simultaneous 95% interval over 31 contrasts. Thick: ordinary 95% interval. Gray band: +/-1 pp research margin.

Primary contrasts with simultaneous uncertainty

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Primary contrasts with simultaneous uncertainty

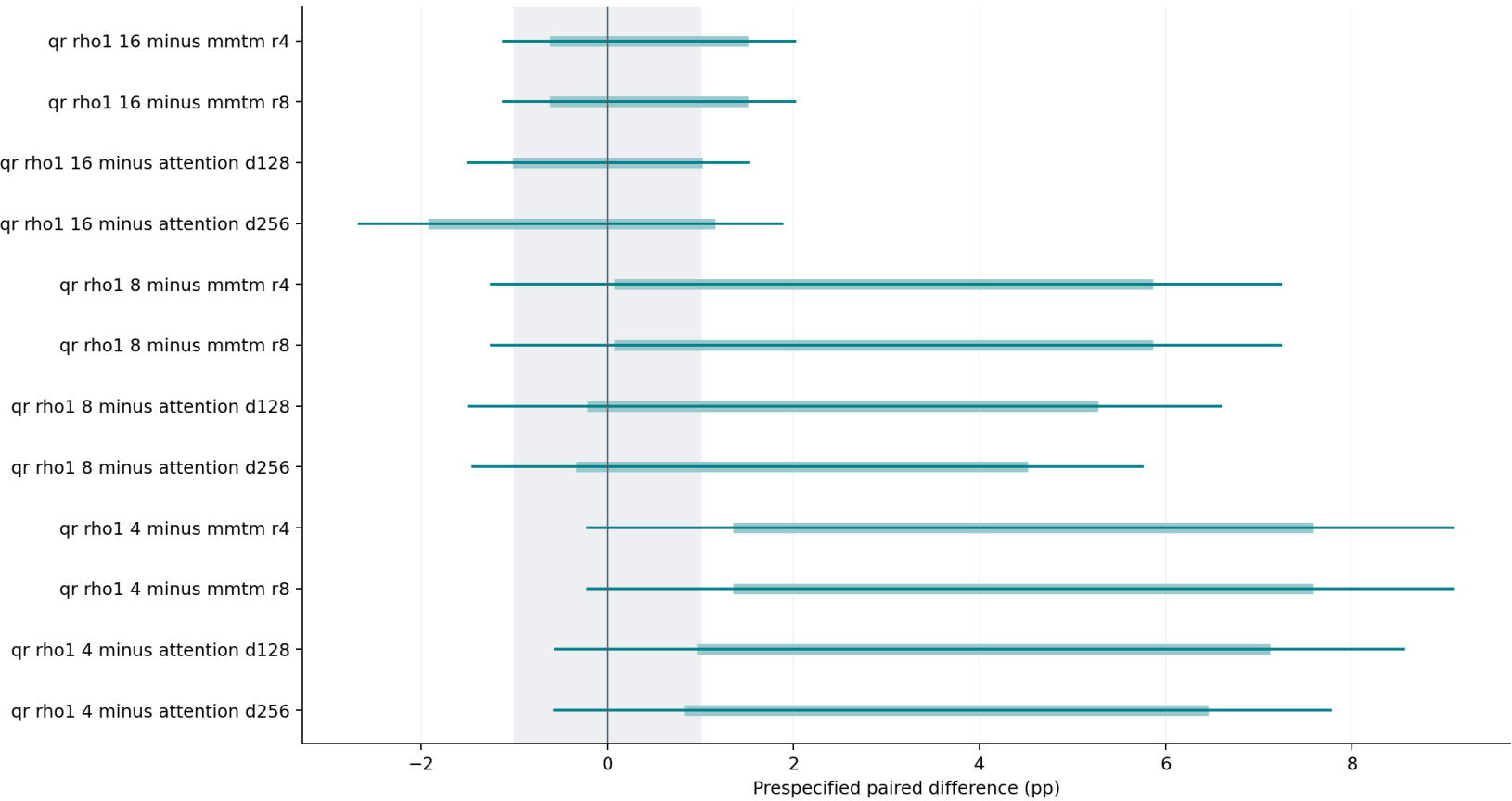


Thin: max-|t| simultaneous 95% interval over 31 contrasts. Thick: ordinary 95% interval. Gray band: +/-1 pp research margin.

Primary contrasts with simultaneous uncertainty

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R&B (Radon Bridge) | Primary contrasts with simultaneous uncertainty

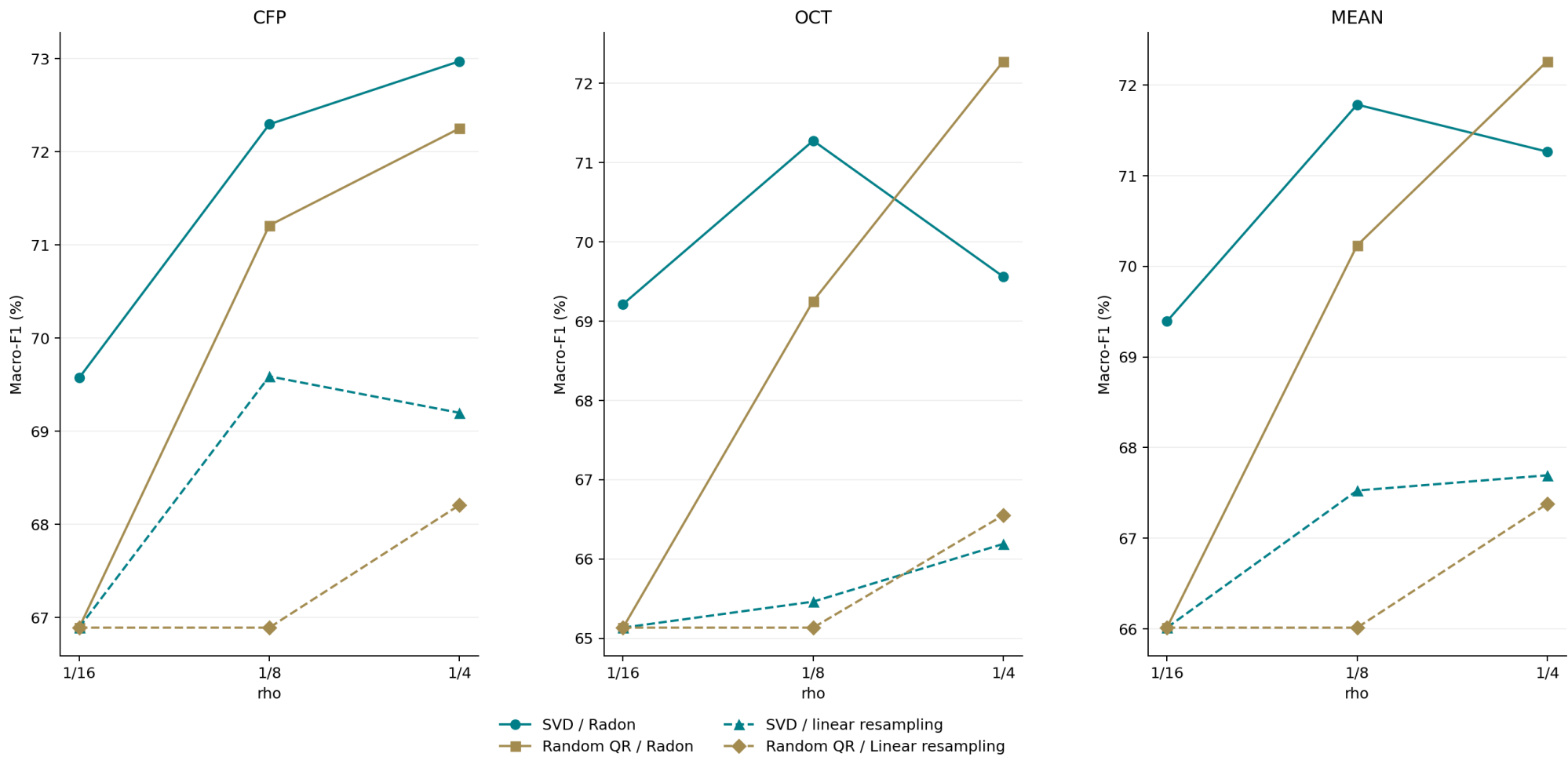


Thin: max-|t| simultaneous 95% interval over 31 contrasts. Thick: ordinary 95% interval. Gray band: +/-1 pp research margin.

Individual training repeat: seed 3416

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Individual training repeat: seed 3416

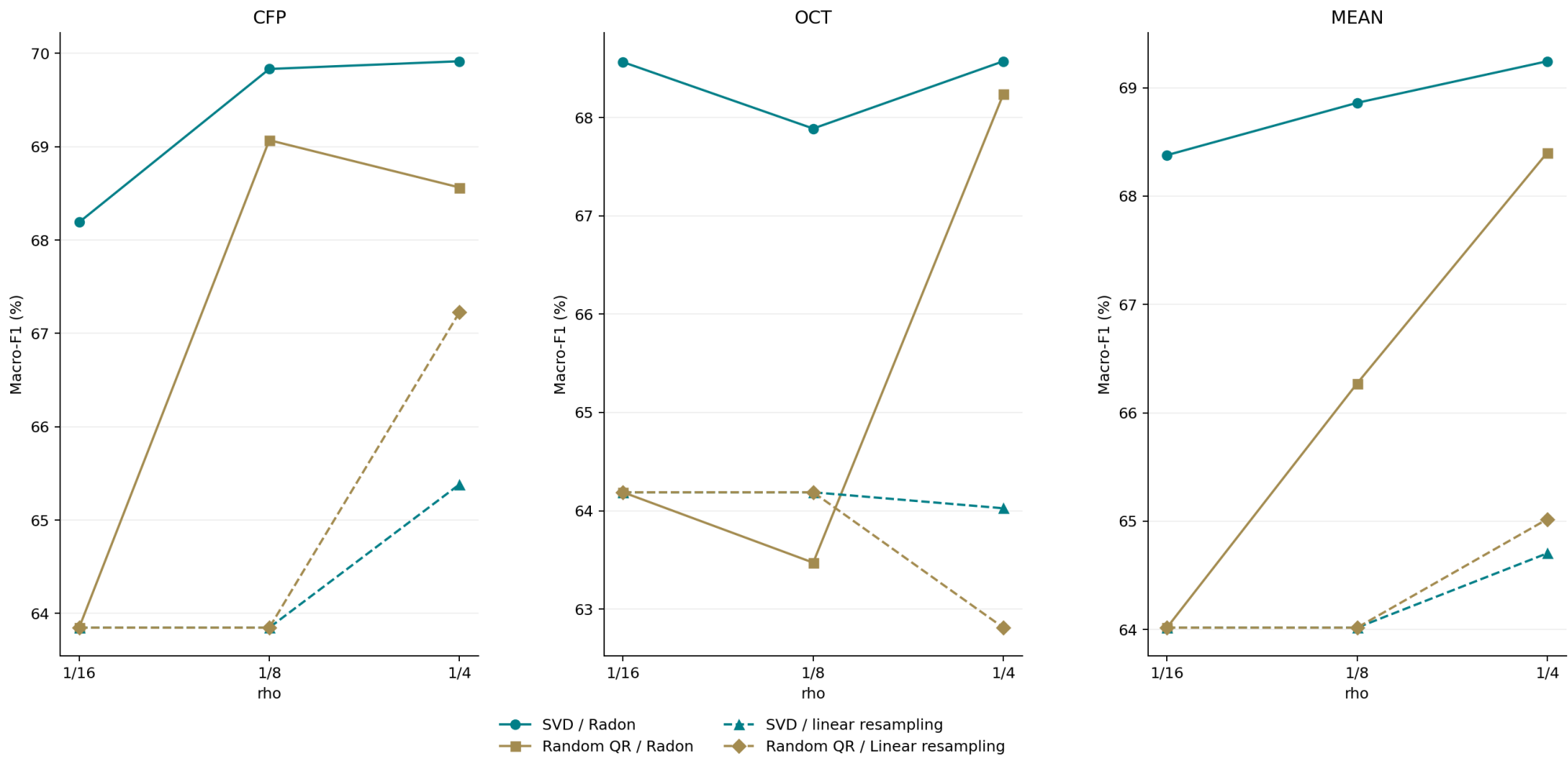


Seeds3416/3417 participated in earlier screening. Seed3418 is separately shown, but uses the same historical development participants.

Individual training repeat: seed 3417

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Individual training repeat: seed 3417

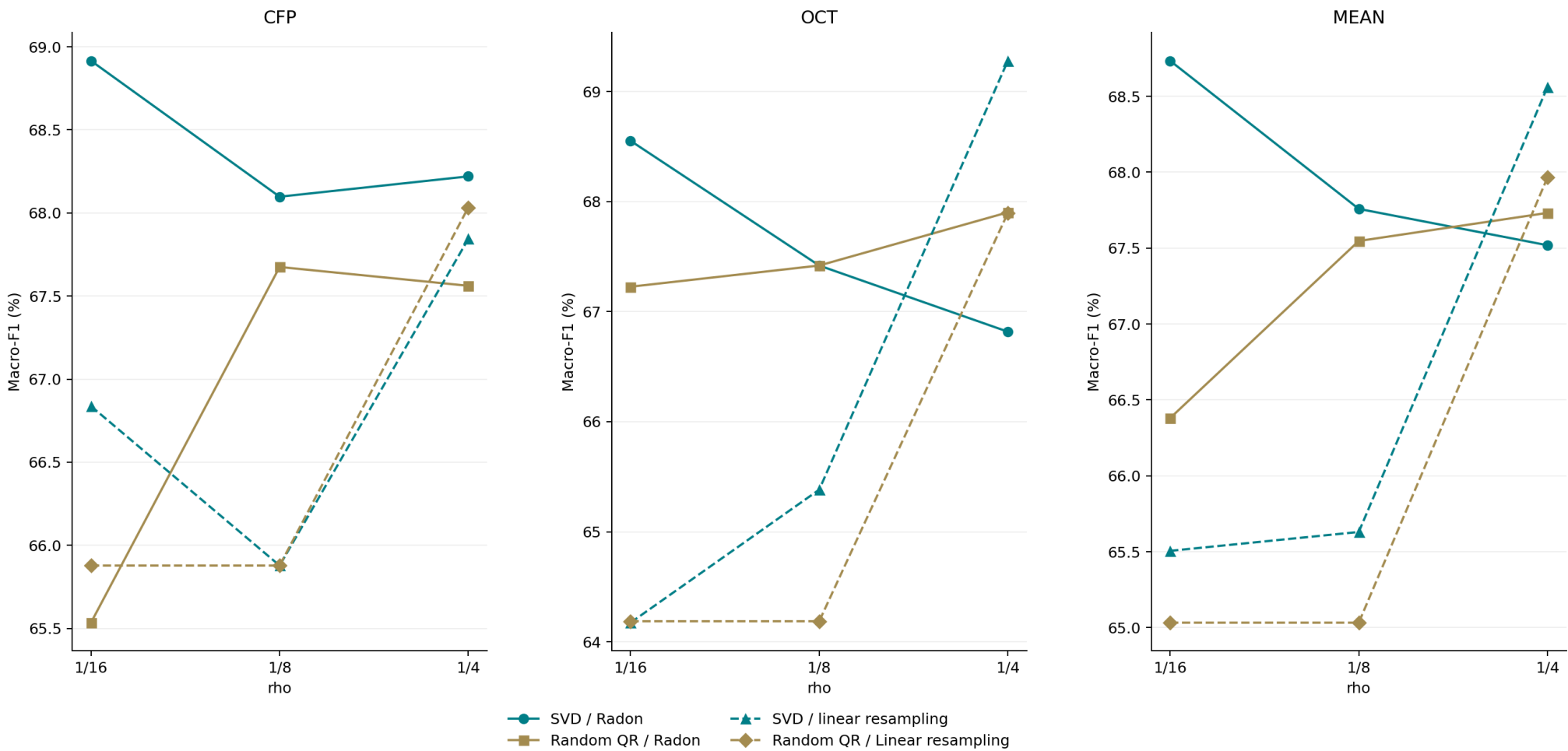


Seeds3416/3417 participated in earlier screening. Seed3418 is separately shown, but uses the same historical development participants.

Individual training repeat: seed 3418

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Individual training repeat: seed 3418

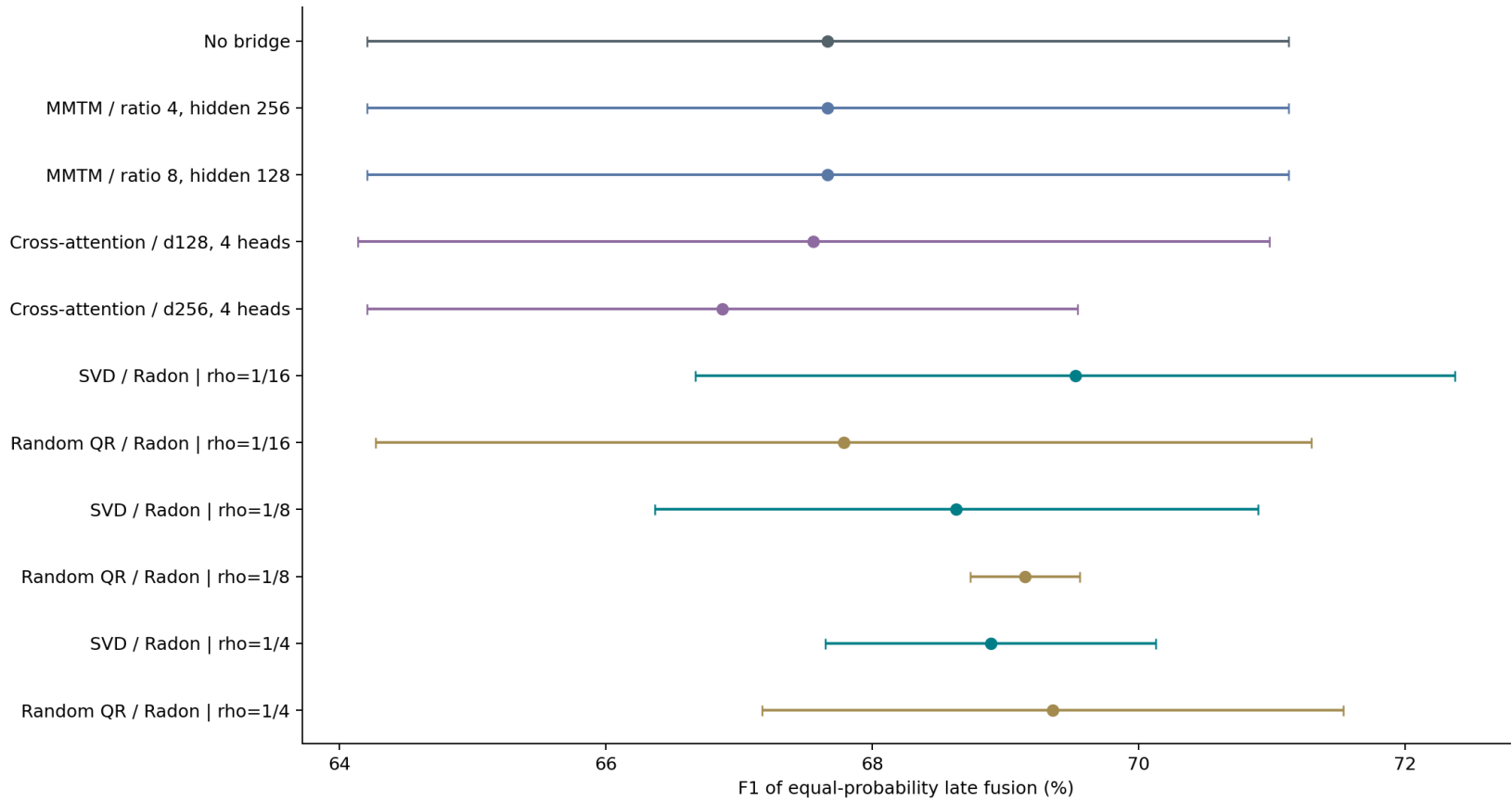


Seeds3416/3417 participated in earlier screening. Seed3418 is separately shown, but uses the same historical development participants.

Late fusion: a separate output

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Late fusion: a separate output

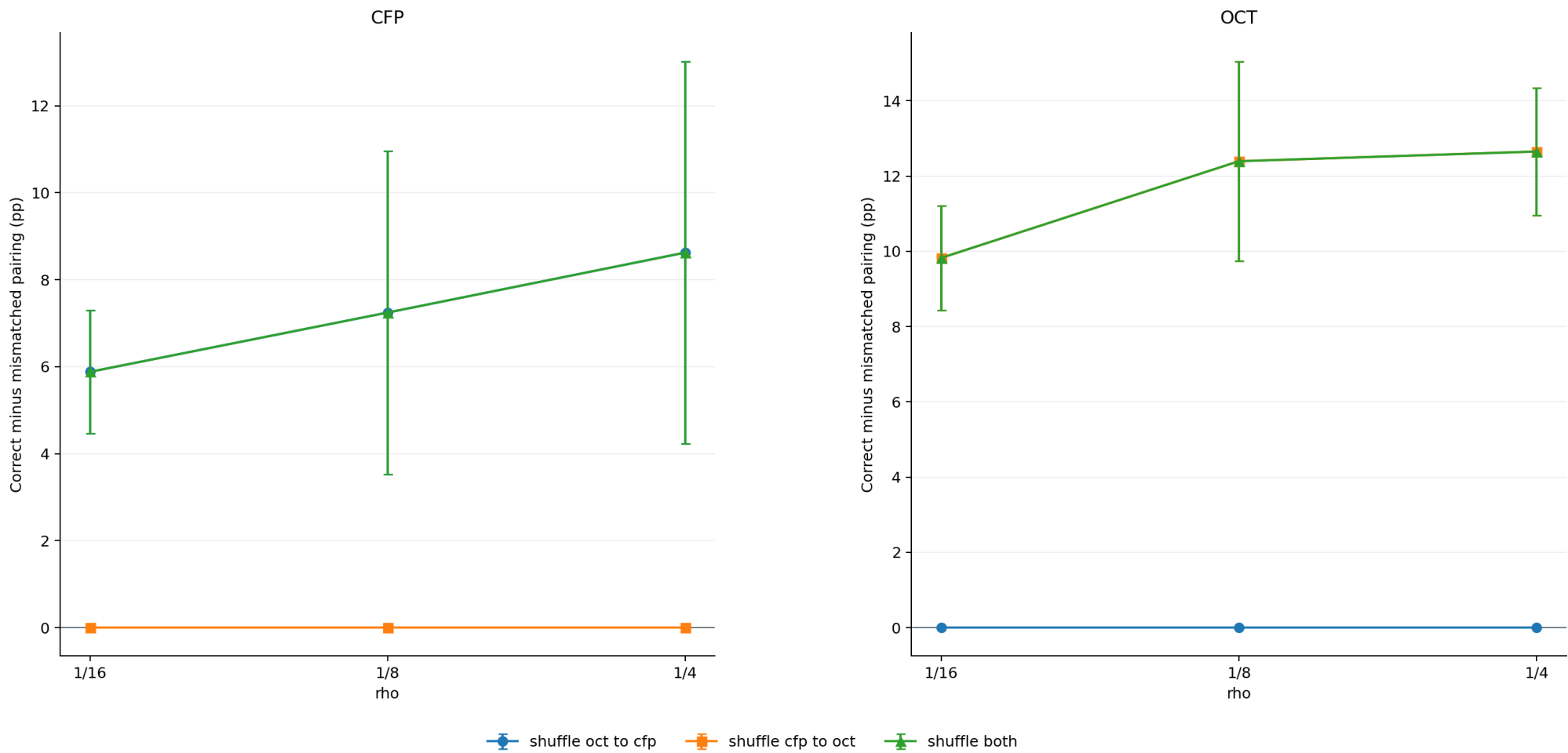


$p_{\text{fused}} = 0.5 p_{\text{CFP}} + 0.5 p_{\text{OCT}}$; fixed argmax; no weight fitting, calibration or threshold search. Secondary analysis.

SVD: use of correctly paired information

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | SVD: use of correctly paired information

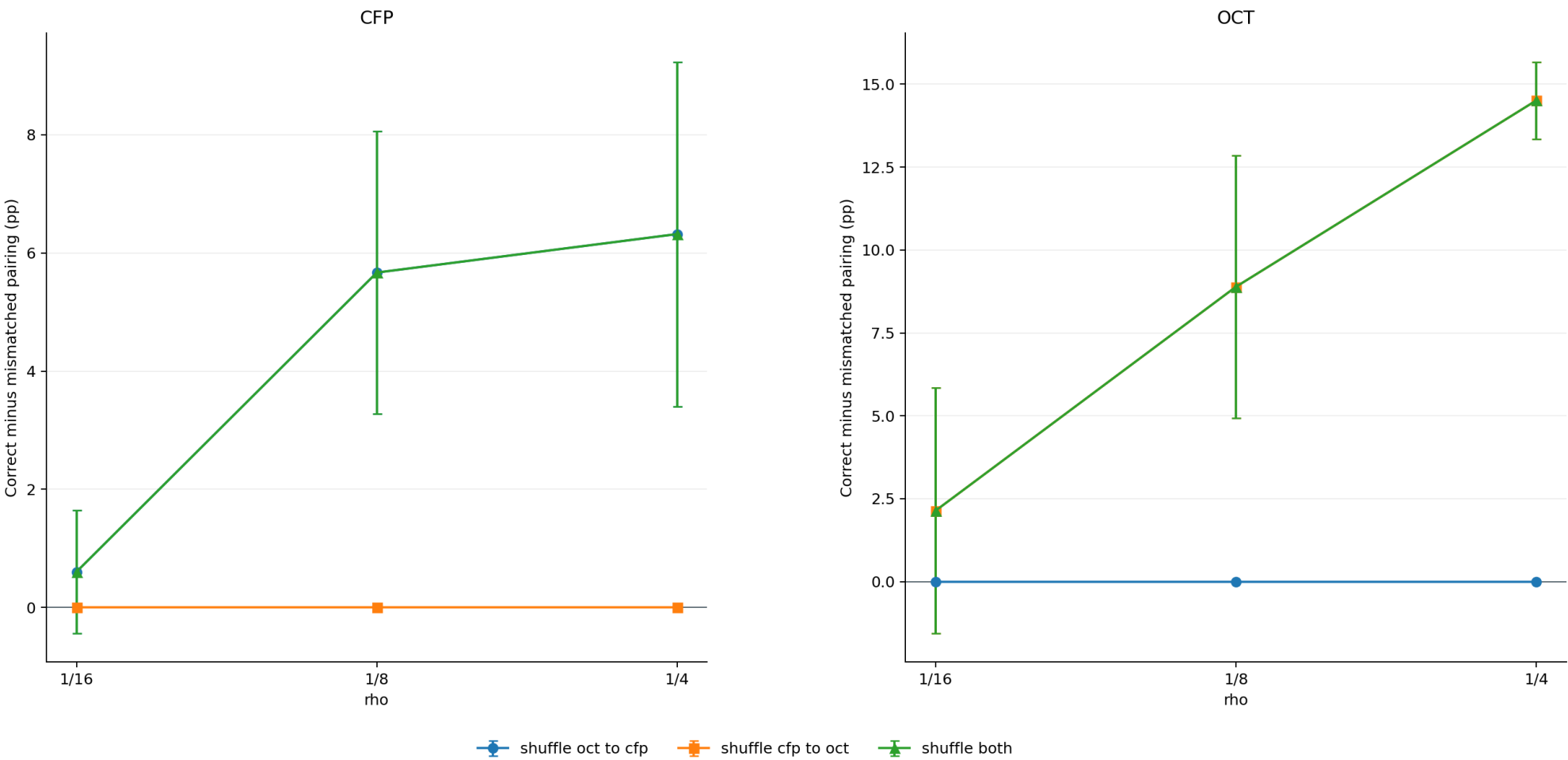


Twenty global derangements per checkpoint, both eyes together. Means over permutations, then seeds. Error bars: seed SD, not extra participants.

QR: use of correctly paired information

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

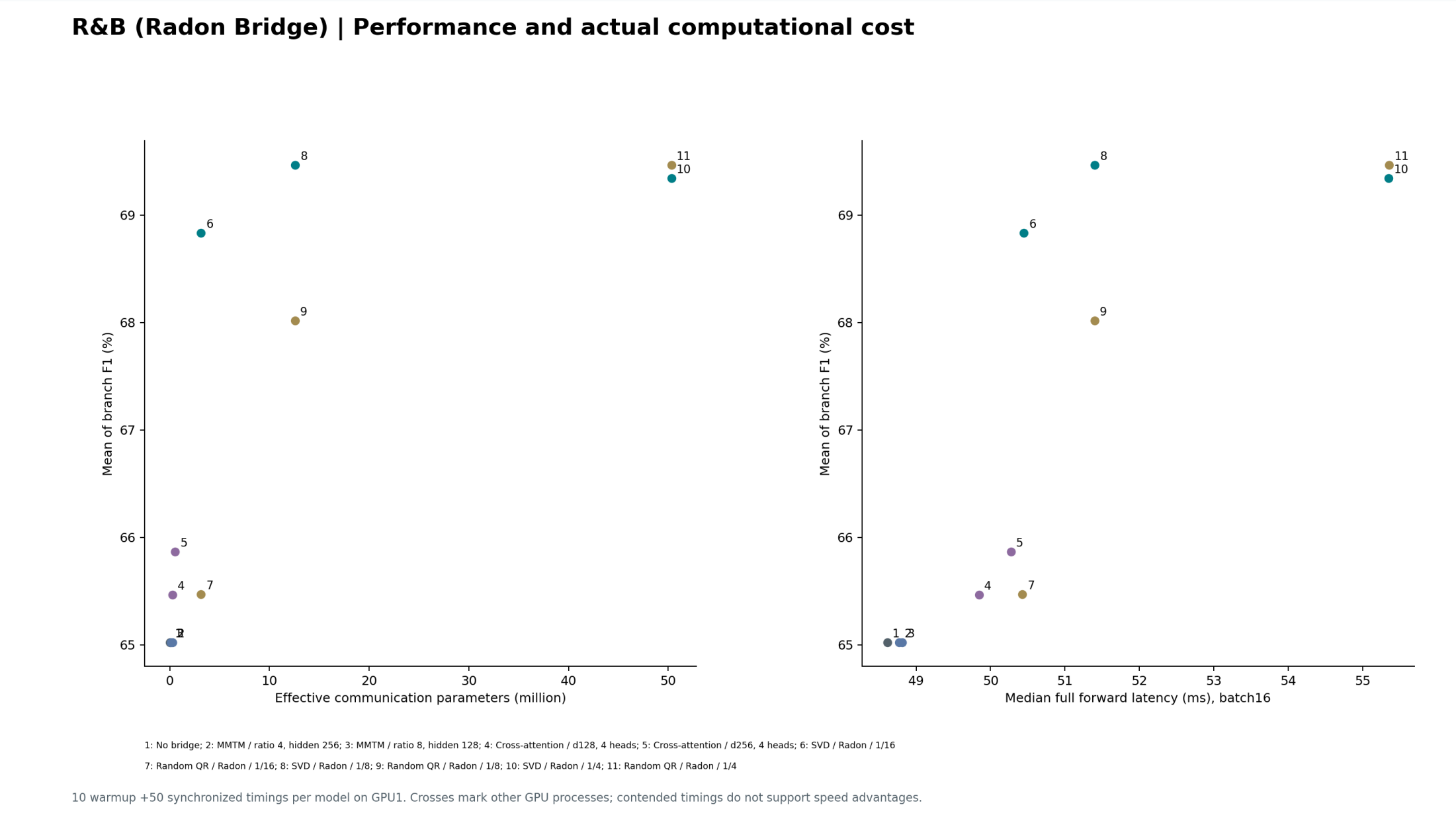
R&B (Radon Bridge) | QR: use of correctly paired information



Twenty global derangements per checkpoint, both eyes together. Means over permutations, then seeds. Error bars: seed SD, not extra participants.

Performance and actual computational cost

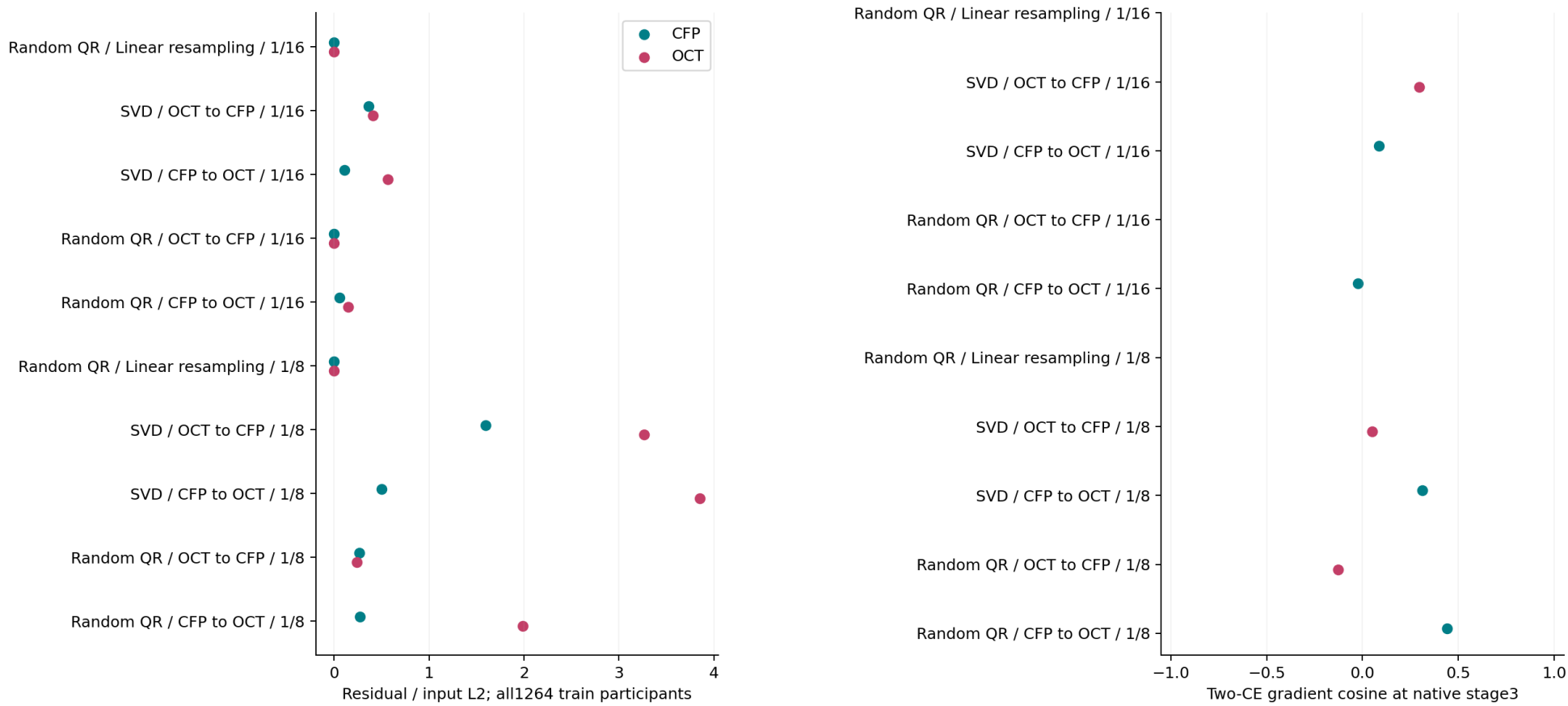
Full vector figure and source data are provided separately; this page preserves the shared plotting convention.



Train-only diagnostics: residuals and task gradients

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Train-only diagnostics: residuals and task gradients

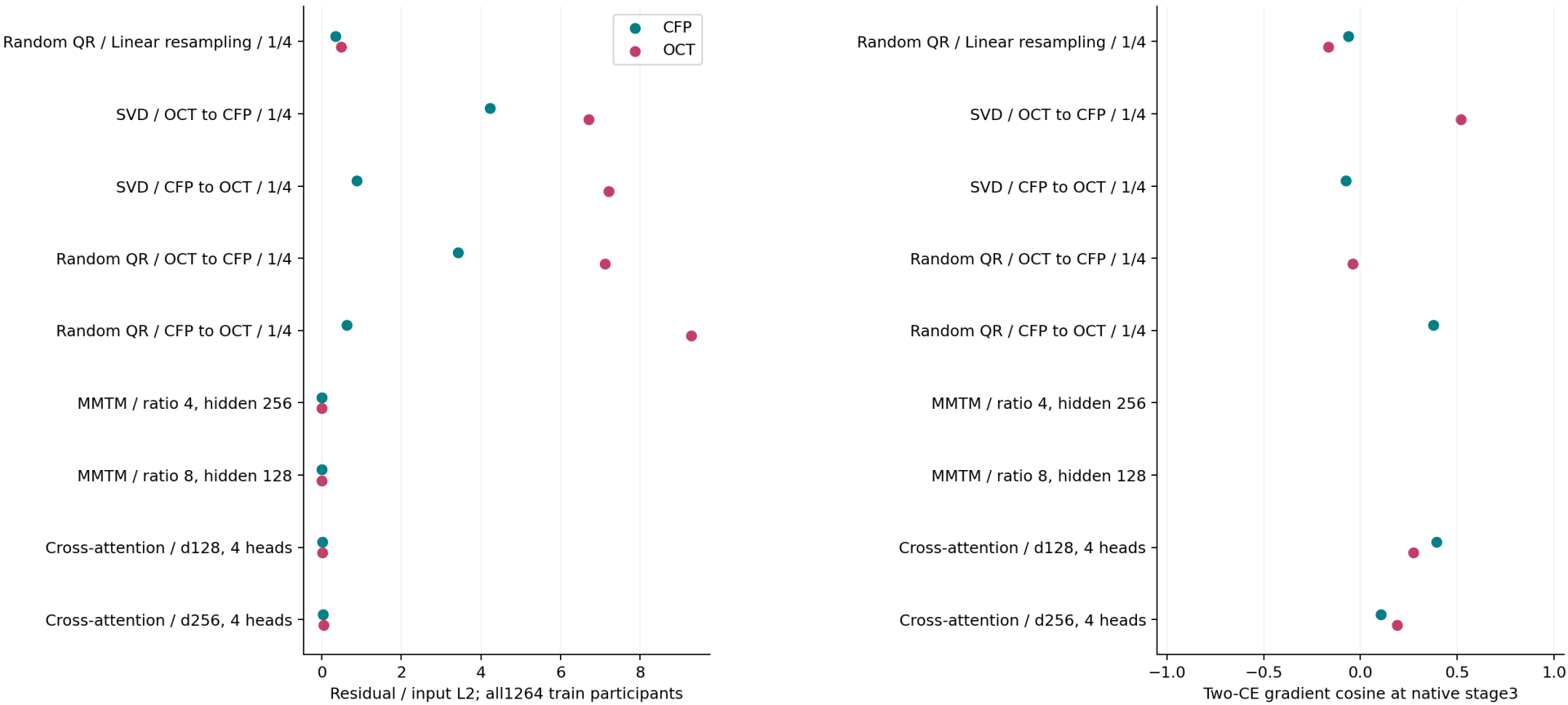


128 fixed training participants for gradients, accumulated before cosine. Undefined gradients omitted; exact per-seed values remain in JSON.

Train-only diagnostics: residuals and task gradients

Full vector figure and source data are provided separately; this page preserves the shared plotting convention.

R&B (Radon Bridge) | Train-only diagnostics: residuals and task gradients



128 fixed training participants for gradients, accumulated before cosine. Undefined gradients omitted; exact per-seed values remain in JSON.

Claim-by-claim evidence and uncertainty

Prespecified primary comparisons 1-6; differences are percentage points.

Contrast	Difference	Simultaneous95% interval	Decision
svd geometry	+3.25	[+0.22, +6.29]	unresolved
qr geometry	+2.04	[-0.51, +4.59]	unresolved
basis geometry interaction	+1.21	[-0.85, +3.27]	unresolved
svd oct to cfp contribution	+4.14	[+0.65, +7.62]	unresolved
svd cfp to oct contribution	+4.46	[-1.29, +10.22]	unresolved
qr oct to cfp contribution	+3.00	[+0.29, +5.71]	unresolved

Claim-by-claim evidence and uncertainty

Prespecified primary comparisons 7-12; differences are percentage points.

Contrast	Difference	Simultaneous95% interval	Decision
qr cfp to oct contribution	+3.18	[-1.54, +7.90]	unresolved
svd rho1 16 minus mmtm r4	+3.81	[-0.04, +7.67]	unresolved
svd rho1 16 minus mmtm r8	+3.81	[-0.04, +7.67]	unresolved
svd rho1 16 minus attention d128	+3.37	[-0.18, +6.92]	unresolved
svd rho1 16 minus attention d256	+2.97	[-0.19, +6.12]	unresolved
svd rho1 8 minus mmtm r4	+4.45	[-0.20, +9.10]	unresolved

Claim-by-claim evidence and uncertainty

Prespecified primary comparisons 13-18; differences are percentage points.

Contrast	Difference	Simultaneous95% interval	Decision
svd rho1 8 minus mmtm r8	+4.45	[-0.20, +9.10]	unresolved
svd rho1 8 minus attention d128	+4.00	[-0.41, +8.41]	unresolved
svd rho1 8 minus attention d256	+3.60	[-0.45, +7.65]	unresolved
svd rho1 4 minus mmtm r4	+4.32	[-0.48, +9.13]	unresolved
svd rho1 4 minus mmtm r8	+4.32	[-0.48, +9.13]	unresolved
svd rho1 4 minus attention d128	+3.88	[-0.79, +8.54]	unresolved

Claim-by-claim evidence and uncertainty

Prespecified primary comparisons 19-24; differences are percentage points.

Contrast	Difference	Simultaneous95% interval	Decision
svd rho1 4 minus attention d256	+3.48	[-0.84, +7.80]	unresolved
qr rho1 16 minus mmtm r4	+0.45	[-1.13, +2.03]	unresolved
qr rho1 16 minus mmtm r8	+0.45	[-1.13, +2.03]	unresolved
qr rho1 16 minus attention d128	+0.00	[-1.51, +1.52]	unresolved
qr rho1 16 minus attention d256	-0.39	[-2.68, +1.89]	unresolved
qr rho1 8 minus mmtm r4	+3.00	[-1.26, +7.25]	unresolved

Claim-by-claim evidence and uncertainty

Prespecified primary comparisons 25-30; differences are percentage points.

Contrast	Difference	Simultaneous95% interval	Decision
qr rho1 8 minus mmtm r8	+3.00	[-1.26, +7.25]	unresolved
qr rho1 8 minus attention d128	+2.55	[-1.51, +6.61]	unresolved
qr rho1 8 minus attention d256	+2.15	[-1.46, +5.77]	unresolved
qr rho1 4 minus mmtm r4	+4.44	[-0.22, +9.11]	unresolved
qr rho1 4 minus mmtm r8	+4.44	[-0.22, +9.11]	unresolved
qr rho1 4 minus attention d128	+4.00	[-0.57, +8.57]	unresolved

Claim-by-claim evidence and uncertainty

Prespecified primary comparisons 31-31; differences are percentage points.

Contrast	Difference	Simultaneous95% interval	Decision
qr rho1 4 minus attention d256	+3.60	[-0.59, +7.79]	unresolved

Direction and correctly paired information

An inference perturbation is a mechanism diagnostic; it is not a randomized clinical trial.

54 selected Radon models: two fixed bases x three directions x three widths x three seeds.

Compare original pairing, each cross direction disabled, both disabled, each direction mismatched, and both mismatched. Self terms, recipient native features and recipient labels remain unchanged.

Twenty label-independent whole-development derangements are shared across models. Both eyes move together; simultaneous mismatch uses a permutation and its inverse.

Cached pre-bridge features are checked against complete MHD probabilities. Show F1 and probability changes, permutation mean/SD/range, and every seed. Diagnostics never affect training or selection.

Data card and biomedical interpretation

UKB paired CFP/OCT; two predictors of the same record-derived glaucoma phenotype.

Data role	Participants	Cases / controls
Training	1264	632 /632
Development	296	148 /148
Test	Not accessed	Not applicable
Reference standard	UKB record-derived clinical phenotype	Not a new expert-image adjudication
Generalizability	Balanced case-control sample	Predictive values/calibration do not imply population performance

Reporting completeness and outstanding author information

TRIPOD-related mapping is an applicability checklist, not a claim of full clinical validation compliance.

Reporting domain	Evidence / location	Remaining need
Participants and source	data_card.json; cache audit; counts and available covariates	Source cohort denominator; complete exclusions
Outcome and predictors	UKB record phenotype; CFP/OCT; documented feature processing	Verify label timing and clinical reference provenance
Model and analysis	Versioned code/configs; initialization; convergence; statistics	No hidden test evaluation claimed
Results and uncertainty	186 rows; all seeds; auxiliary metrics;31 simultaneous intervals	Independent external validation
Ethics and availability	Restricted participant-level evidence retained in authorized environment	Confirm ethics, UKB application/license and sharing terms

What would strengthen a future NBE submission?

External evidence should answer new questions; more variants on this development set cannot replace it.

Independent data: prespecify the model and evaluation, then validate on an appropriate external or independently held-out cohort.

Generalization: extend to additional clinically meaningful tasks, modalities and backbone families under separate protocols.

Biomedical contribution: establish the practical setting in which connecting existing task networks improves a useful clinical or scientific outcome.

Strong task-specific comparators and realistic prevalence: assess reliability, subgroup effects, missing modalities and actual deployment constraints where relevant.

If a primary interval is unresolved, say so. A more complex bridge or a larger number of in-sample ablations is not an automatic remedy.

Sources, reproducibility and evidence boundaries

Full code, protocol, environment, aggregate data and hashes accompany this report.

MMTM: Joze et al., CVPR2020. Paper2*sigmoid excitation; this adapter adds zero output initialization. Public author code differs in its excitation scale.

Attention: Vaswani et al., NeurIPS2017. Standard scaled dot-product multi-head attention adapted for bidirectional stage3 communication; no complete Transformer claim.

NBE: aims and scope, reporting standards, and clinical research policies. Relevant reporting domains are mapped; unverified ethics/license details remain explicit.

Participant-level predictions, features and permutations stay in the authorized environment. GitHub contains aggregate evidence and reproducible source.

Source commit: f850929b57dfb5bece1805d63f65da7f0cc0daf8

Complete second-stage results

Accepted results 1-12 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3416	No bridge / N/A	66.89	65.14	66.01	70.59	0/8
3416	Learned CM / 1/16	70.22	70.18	70.20	70.55	11/17
3416	SVD / Radon / 1/16	69.57	69.21	69.39	70.24	9/15
3416	SVD / self only / 1/16	66.89	65.14	66.01	70.59	0/8
3416	SVD / scrambled Radon / 1/16	68.10	65.71	66.90	68.42	7/13
3416	SVD / linear resampling / 1/16	66.89	65.14	66.01	70.59	0/8
3416	Random QR / Radon / 1/16	66.89	65.14	66.01	70.59	0/8
3416	Learned CM / 1/8	69.64	70.53	70.09	71.43	16/22
3416	SVD / Radon / 1/8	72.30	71.28	71.79	70.94	11/17
3416	SVD / self only / 1/8	66.89	65.14	66.01	70.59	0/8
3416	SVD / scrambled Radon / 1/8	68.24	69.21	68.73	71.59	10/10
3416	SVD / linear resampling / 1/8	69.59	65.46	67.53	67.84	9/15

Complete second-stage results

Accepted results 13-24 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3416	Random QR / Radon / 1/8	71.21	69.25	70.23	69.53	7/13
3416	Learned CM / 1/4	69.90	70.20	70.05	69.89	10/16
3416	SVD / Radon / 1/4	72.97	69.56	71.27	70.26	9/15
3416	SVD / self only / 1/4	66.89	65.14	66.01	70.59	0/8
3416	SVD / scrambled Radon / 1/4	68.48	70.51	69.49	70.20	14/20
3416	SVD / linear resampling / 1/4	69.20	66.19	67.69	70.24	11/17
3416	Random QR / Radon / 1/4	72.25	72.28	72.26	71.59	9/15
3417	No bridge / N/A	63.85	64.19	64.02	63.85	0/8
3417	Learned CM / 1/16	66.89	65.77	66.33	69.16	7/13
3417	SVD / Radon / 1/16	68.19	68.56	68.38	71.95	16/22
3417	SVD / self only / 1/16	63.85	64.19	64.02	63.85	0/8
3417	SVD / scrambled Radon / 1/16	67.89	63.50	65.69	68.91	11/17

Complete second-stage results

Accepted results 25-36 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3417	SVD / linear resampling / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Random QR / Radon / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Learned CM / 1/8	65.88	65.83	65.85	65.53	10/13
3417	SVD / Radon / 1/8	69.83	67.89	68.86	68.54	10/16
3417	SVD / self only / 1/8	63.85	64.19	64.02	63.85	0/8
3417	SVD / scrambled Radon / 1/8	65.54	67.91	66.72	67.56	10/16
3417	SVD / linear resampling / 1/8	63.85	64.19	64.02	63.85	0/8
3417	Random QR / Radon / 1/8	69.07	63.47	66.27	69.20	6/12
3417	Learned CM / 1/4	68.91	69.93	69.42	69.93	10/16
3417	SVD / Radon / 1/4	69.92	68.57	69.24	68.56	12/18
3417	SVD / self only / 1/4	63.85	64.19	64.02	63.85	0/8
3417	SVD / scrambled Radon / 1/4	68.17	67.47	67.82	67.15	11/17

Complete second-stage results

Accepted results 37-48 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3417	SVD / linear resampling / 1/4	65.38	64.02	64.70	65.67	6/12
3417	Random QR / Radon / 1/4	68.56	68.24	68.40	69.24	10/16
3418	No bridge / N/A	65.88	64.19	65.03	68.55	0/8
3418	Learned CM / 1/16	68.48	68.10	68.29	67.47	8/14
3418	SVD / Radon / 1/16	68.92	68.55	68.73	66.38	14/20
3418	SVD / self only / 1/16	65.88	64.19	65.03	68.55	0/8
3418	SVD / scrambled Radon / 1/16	65.81	65.95	65.88	68.92	5/11
3418	SVD / linear resampling / 1/16	66.84	64.17	65.51	69.24	10/14
3418	Random QR / Radon / 1/16	65.53	67.23	66.38	68.92	9/15
3418	Learned CM / 1/8	65.88	64.19	65.03	68.55	0/8
3418	SVD / Radon / 1/8	68.10	67.42	67.76	66.42	14/20
3418	SVD / self only / 1/8	65.88	64.19	65.03	68.55	0/8

Complete second-stage results

Accepted results 49-60 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3418	SVD / scrambled Radon / 1/8	65.20	68.19	66.70	69.25	9/13
3418	SVD / linear resampling / 1/8	65.88	65.38	65.63	67.17	12/12
3418	Random QR / Radon / 1/8	67.67	67.42	67.55	68.71	14/20
3418	Learned CM / 1/4	65.88	64.19	65.03	68.55	0/8
3418	SVD / Radon / 1/4	68.22	66.82	67.52	67.84	6/12
3418	SVD / self only / 1/4	65.88	64.19	65.03	68.55	0/8
3418	SVD / scrambled Radon / 1/4	67.57	66.88	67.22	66.22	9/15
3418	SVD / linear resampling / 1/4	67.84	69.28	68.56	70.72	5/11
3418	Random QR / Radon / 1/4	67.56	67.90	67.73	67.23	12/18
3416	Centered-fit SVD / Radon / 1/16	67.53	67.88	67.70	69.93	8/14
3416	Centered-fit SVD / self only / 1/16	66.89	65.14	66.01	70.59	0/8
3416	Centered-fit SVD / scrambled Radon / 1/16	69.26	68.10	68.68	67.86	10/16

Complete second-stage results

Accepted results 61-72 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3416	Centered-fit SVD / linear resampling / 1/16	66.89	65.14	66.01	70.59	0/8
3416	Centered-fit SVD / Radon / 1/8	74.61	73.30	73.96	73.98	28/28
3416	Centered-fit SVD / self only / 1/8	66.89	65.14	66.01	70.59	0/8
3416	Centered-fit SVD / scrambled Radon / 1/8	70.94	69.18	70.06	68.85	8/14
3416	Centered-fit SVD / linear resampling / 1/8	66.89	65.14	66.01	70.59	0/8
3416	Centered-fit SVD / Radon / 1/4	73.59	73.53	73.56	72.85	7/13
3416	Centered-fit SVD / self only / 1/4	66.89	65.14	66.01	70.59	0/8
3416	Centered-fit SVD / scrambled Radon / 1/4	73.59	70.59	72.09	71.93	7/13
3416	Centered-fit SVD / linear resampling / 1/4	69.57	67.82	68.70	72.29	5/11
3417	Centered-fit SVD / Radon / 1/16	68.75	66.87	67.81	71.84	18/24
3417	Centered-fit SVD / self only / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Centered-fit SVD / scrambled Radon / 1/16	64.53	67.57	66.05	67.88	11/17

Complete second-stage results

Accepted results 73-84 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3417	Centered-fit SVD / linear resampling / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Centered-fit SVD / Radon / 1/8	68.87	68.57	68.72	68.91	9/15
3417	Centered-fit SVD / self only / 1/8	63.85	64.19	64.02	63.85	0/8
3417	Centered-fit SVD / scrambled Radon / 1/8	67.57	68.55	68.06	68.22	11/17
3417	Centered-fit SVD / linear resampling / 1/8	64.44	64.02	64.23	65.31	6/12
3417	Centered-fit SVD / Radon / 1/4	67.77	69.57	68.67	67.84	8/14
3417	Centered-fit SVD / self only / 1/4	63.85	64.19	64.02	63.85	0/8
3417	Centered-fit SVD / scrambled Radon / 1/4	68.91	62.30	65.61	67.12	6/12
3417	Centered-fit SVD / linear resampling / 1/4	67.18	63.17	65.18	67.21	8/14
3418	Centered-fit SVD / Radon / 1/16	68.55	67.84	68.20	69.53	17/23
3418	Centered-fit SVD / self only / 1/16	65.88	64.19	65.03	68.55	0/8
3418	Centered-fit SVD / scrambled Radon / 1/16	68.91	65.87	67.39	69.25	13/19

Complete second-stage results

Accepted results 85-96 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3418	Centered-fit SVD / linear resampling / 1/16	65.88	64.19	65.03	68.55	0/8
3418	Centered-fit SVD / Radon / 1/8	67.82	64.79	66.30	69.26	3/9
3418	Centered-fit SVD / self only / 1/8	65.88	64.19	65.03	68.55	0/8
3418	Centered-fit SVD / scrambled Radon / 1/8	65.03	66.79	65.91	66.47	12/18
3418	Centered-fit SVD / linear resampling / 1/8	65.88	64.19	65.03	68.55	0/8
3418	Centered-fit SVD / Radon / 1/4	67.89	65.85	66.87	67.22	7/13
3418	Centered-fit SVD / self only / 1/4	65.88	64.19	65.03	68.55	0/8
3418	Centered-fit SVD / scrambled Radon / 1/4	69.54	67.82	68.68	67.82	6/12
3418	Centered-fit SVD / linear resampling / 1/4	65.83	67.77	66.80	65.41	8/14
3416	Learned channel / Radon / 1/16	67.42	67.09	67.26	68.42	7/13
3416	Learned channel / Self only / 1/16	66.89	65.14	66.01	70.59	0/8
3416	Learned channel / Scrambled Radon / 1/16	68.15	65.52	66.83	67.82	7/13

Complete second-stage results

Accepted results 97-108 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3416	Learned channel / Linear resampling / 1/16	66.89	65.14	66.01	70.59	0/8
3416	Learned channel / Radon / 1/8	70.92	68.24	69.58	68.91	10/16
3416	Learned channel / Self only / 1/8	66.89	65.14	66.01	70.59	0/8
3416	Learned channel / Scrambled Radon / 1/8	71.61	67.77	69.69	72.29	10/16
3416	Learned channel / Linear resampling / 1/8	66.89	65.14	66.01	70.59	0/8
3416	Learned channel / Radon / 1/4	71.27	72.62	71.94	70.91	11/17
3416	Learned channel / Self only / 1/4	66.89	65.14	66.01	70.59	0/8
3416	Learned channel / Scrambled Radon / 1/4	68.17	70.16	69.17	70.61	9/13
3416	Learned channel / Linear resampling / 1/4	69.58	67.90	68.74	70.93	15/21
3417	Learned channel / Radon / 1/16	65.79	63.34	64.56	66.91	6/12
3417	Learned channel / Self only / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Learned channel / Scrambled Radon / 1/16	63.85	64.19	64.02	63.85	0/8

Complete second-stage results

Accepted results 109-120 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3417	Learned channel / Linear resampling / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Learned channel / Radon / 1/8	67.47	68.21	67.84	68.24	6/12
3417	Learned channel / Self only / 1/8	63.85	64.19	64.02	63.85	0/8
3417	Learned channel / Scrambled Radon / 1/8	65.54	63.84	64.69	68.22	10/16
3417	Learned channel / Linear resampling / 1/8	63.85	64.19	64.02	63.85	0/8
3417	Learned channel / Radon / 1/4	68.58	67.54	68.06	68.57	11/17
3417	Learned channel / Self only / 1/4	63.85	64.19	64.02	63.85	0/8
3417	Learned channel / Scrambled Radon / 1/4	69.13	65.17	67.15	66.82	8/14
3417	Learned channel / Linear resampling / 1/4	63.85	64.19	64.02	63.85	0/8
3418	Learned channel / Radon / 1/16	66.89	68.23	67.56	69.54	12/18
3418	Learned channel / Self only / 1/16	65.88	64.19	65.03	68.55	0/8
3418	Learned channel / Scrambled Radon / 1/16	65.88	64.19	65.03	68.55	0/8

Complete second-stage results

Accepted results 121-132 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3418	Learned channel / Linear resampling / 1/16	65.88	64.19	65.03	68.55	0/8
3418	Learned channel / Radon / 1/8	66.55	67.91	67.23	66.55	7/13
3418	Learned channel / Self only / 1/8	65.88	64.19	65.03	68.55	0/8
3418	Learned channel / Scrambled Radon / 1/8	67.57	66.55	67.06	69.25	10/16
3418	Learned channel / Linear resampling / 1/8	68.54	64.17	66.36	69.93	7/13
3418	Learned channel / Radon / 1/4	68.88	65.54	67.21	67.91	17/23
3418	Learned channel / Self only / 1/4	62.75	69.25	66.00	67.47	4/10
3418	Learned channel / Scrambled Radon / 1/4	68.58	67.86	68.22	67.23	4/10
3418	Learned channel / Linear resampling / 1/4	66.89	67.55	67.22	69.93	13/18
3416	Random QR / Linear resampling / 1/16	66.89	65.14	66.01	70.59	0/8
3416	SVD / OCT to CFP / 1/16	70.13	63.16	66.64	70.90	14/14
3416	SVD / CFP to OCT / 1/16	68.22	66.54	67.38	70.94	9/15

Complete second-stage results

Accepted results 133-144 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3416	Random QR / OCT to CFP / 1/16	66.89	65.14	66.01	70.59	0/8
3416	Random QR / CFP to OCT / 1/16	66.21	67.90	67.06	70.55	8/13
3416	Random QR / Linear resampling / 1/8	66.89	65.14	66.01	70.59	0/8
3416	SVD / OCT to CFP / 1/8	71.62	63.83	67.73	68.24	12/18
3416	SVD / CFP to OCT / 1/8	65.63	70.94	68.29	69.10	4/10
3416	Random QR / OCT to CFP / 1/8	66.89	65.14	66.01	70.59	0/8
3416	Random QR / CFP to OCT / 1/8	66.54	66.89	66.72	70.93	9/14
3416	Random QR / Linear resampling / 1/4	68.21	66.55	67.38	69.93	9/15
3416	SVD / OCT to CFP / 1/4	70.07	67.77	68.92	70.53	24/25
3416	SVD / CFP to OCT / 1/4	68.91	69.24	69.08	67.90	11/17
3416	Random QR / OCT to CFP / 1/4	72.93	63.27	68.10	70.86	10/14
3416	Random QR / CFP to OCT / 1/4	65.53	68.56	67.04	67.56	12/18

Complete second-stage results

Accepted results 145-156 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3417	Random QR / Linear resampling / 1/16	63.85	64.19	64.02	63.85	0/8
3417	SVD / OCT to CFP / 1/16	65.87	62.82	64.35	66.89	11/13
3417	SVD / CFP to OCT / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Random QR / OCT to CFP / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Random QR / CFP to OCT / 1/16	63.85	64.19	64.02	63.85	0/8
3417	Random QR / Linear resampling / 1/8	63.85	64.19	64.02	63.85	0/8
3417	SVD / OCT to CFP / 1/8	67.91	64.19	66.05	67.54	8/14
3417	SVD / CFP to OCT / 1/8	63.85	64.19	64.02	63.85	0/8
3417	Random QR / OCT to CFP / 1/8	67.82	62.08	64.95	69.16	8/14
3417	Random QR / CFP to OCT / 1/8	63.85	64.19	64.02	63.85	0/8
3417	Random QR / Linear resampling / 1/4	67.23	62.81	65.02	66.85	6/12
3417	SVD / OCT to CFP / 1/4	70.25	62.50	66.37	68.21	8/12

Complete second-stage results

Accepted results 157-168 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3417	SVD / CFP to OCT / 1/4	63.14	66.51	64.82	64.53	10/16
3417	Random QR / OCT to CFP / 1/4	71.62	65.20	68.41	66.55	17/23
3417	Random QR / CFP to OCT / 1/4	61.79	68.85	65.32	65.14	7/13
3418	Random QR / Linear resampling / 1/16	65.88	64.19	65.03	68.55	0/8
3418	SVD / OCT to CFP / 1/16	65.88	64.19	65.03	68.55	0/8
3418	SVD / CFP to OCT / 1/16	65.88	64.19	65.03	68.55	0/8
3418	Random QR / OCT to CFP / 1/16	65.88	64.19	65.03	68.55	0/8
3418	Random QR / CFP to OCT / 1/16	65.88	64.19	65.03	68.55	0/8
3418	Random QR / Linear resampling / 1/8	65.88	64.19	65.03	68.55	0/8
3418	SVD / OCT to CFP / 1/8	63.95	66.54	65.24	65.38	4/10
3418	SVD / CFP to OCT / 1/8	65.44	67.18	66.31	67.23	4/10
3418	Random QR / OCT to CFP / 1/8	65.88	64.19	65.03	68.55	0/8

Complete second-stage results

Accepted results 169-180 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3418	Random QR / CFP to OCT / 1/8	66.09	69.54	67.82	67.84	12/18
3418	Random QR / Linear resampling / 1/4	68.03	67.90	67.96	70.11	5/11
3418	SVD / OCT to CFP / 1/4	70.86	62.70	66.78	68.80	8/14
3418	SVD / CFP to OCT / 1/4	65.88	64.19	65.03	68.55	0/8
3418	Random QR / OCT to CFP / 1/4	70.61	63.12	66.86	71.28	5/11
3418	Random QR / CFP to OCT / 1/4	65.88	64.19	65.03	68.55	0/8
3416	MMTM / ratio 4, hidden 256 / N/A	66.89	65.14	66.01	70.59	0/8
3416	MMTM / ratio 8, hidden 128 / N/A	66.89	65.14	66.01	70.59	0/8
3416	Cross-attention / d128, 4 heads / N/A	66.89	65.14	66.01	70.59	0/8
3416	Cross-attention / d256, 4 heads / N/A	67.86	65.20	66.53	67.89	7/13
3417	MMTM / ratio 4, hidden 256 / N/A	63.85	64.19	64.02	63.85	0/8
3417	MMTM / ratio 8, hidden 128 / N/A	63.85	64.19	64.02	63.85	0/8

Complete second-stage results

Accepted results 181-186 of186. F1 in percent; fusion is a separate output.

Seed	Method / rho	CFP	OCT	Mean	Fusion	Best/stop
3417	Cross-attention / d128, 4 heads / N/A	63.85	64.19	64.02	63.85	0/8
3417	Cross-attention / d256, 4 heads / N/A	63.85	64.19	64.02	63.85	0/8
3418	MMTM / ratio 4, hidden 256 / N/A	65.88	64.19	65.03	68.55	0/8
3418	MMTM / ratio 8, hidden 128 / N/A	65.88	64.19	65.03	68.55	0/8
3418	Cross-attention / d128, 4 heads / N/A	64.84	67.90	66.37	68.24	10/16
3418	Cross-attention / d256, 4 heads / N/A	64.19	69.90	67.05	68.88	17/18

Two follow-ups, separate research questions

18 QR controls +9 joint-width trainings; same parents, bases, seeds and development participants.

A tests whether cross-source and spatial-structure explanations extend to random QR. Self-only and scrambled Radon use the same per-seed QR basis as standard Radon.

B tests a new training regime: three nested widths on the same 16 participants, equal mean losses, one optimizer update; the six branch/width F1 values choose one checkpoint.

Shared BN starts each width from the same pre-batch buffers; running-stat updates are averaged and the batch counter advances once. This is not pooled-mixture variance.

B changes parameter sharing, gradient sources, BN updates, checkpoint selection and work per epoch. A failure of this regime does not prove that every multi-width training algorithm fails.

No new pretraining, basis fitting, data, task, backbone or width search. Native parameters and BN remain trainable. Original convergence and memory requirements are preserved.

Study A: communication survives; geometry is basis-dependent

QR Radon, QR self-only and QR scrambling are complete across all nine seed/width cells.

QR Radon - self-only: **+2.63 pp**; simultaneous 95% CI [+0.24, +5.02]. All 9 QR self controls select epoch 0. This supports a positive cross-source contribution under the current protocol.

QR Radon - scrambled: **+0.18 pp**; simultaneous 95% CI [-1.12, +1.48]. Current evidence resolves neither a useful geometric advantage nor practical equivalence.

Noncentered-SVD Radon exceeds its scrambled counterpart in 9/9 cells; average gap 1.98 pp. The SVD-minus-QR spatial-gain interaction is +1.79 pp, simultaneous CI [+0.03, +3.56].

A stronger SVD spatial gain does not imply an independent semantic alignment or a universal advantage of Radon geometry. The basis and spatial operator act together.

The positive QR-self and interaction intervals do not lie entirely above +1 pp; they remain unresolved under the stricter substantive-effect classification.

Study B: flat width curves are a negative result

All nine shared models select epoch0; all27 exported views recover the native baseline predictions.

Every joint-width training ran8 epochs and met the unchanged plateau rule. It did not improve the common development selection score above initialization.

Mean branch F1 is65.02% across all three compression families. Relative to independent widths: learned channel -2.90 pp; SVD -4.19 pp; QR -2.63 pp.

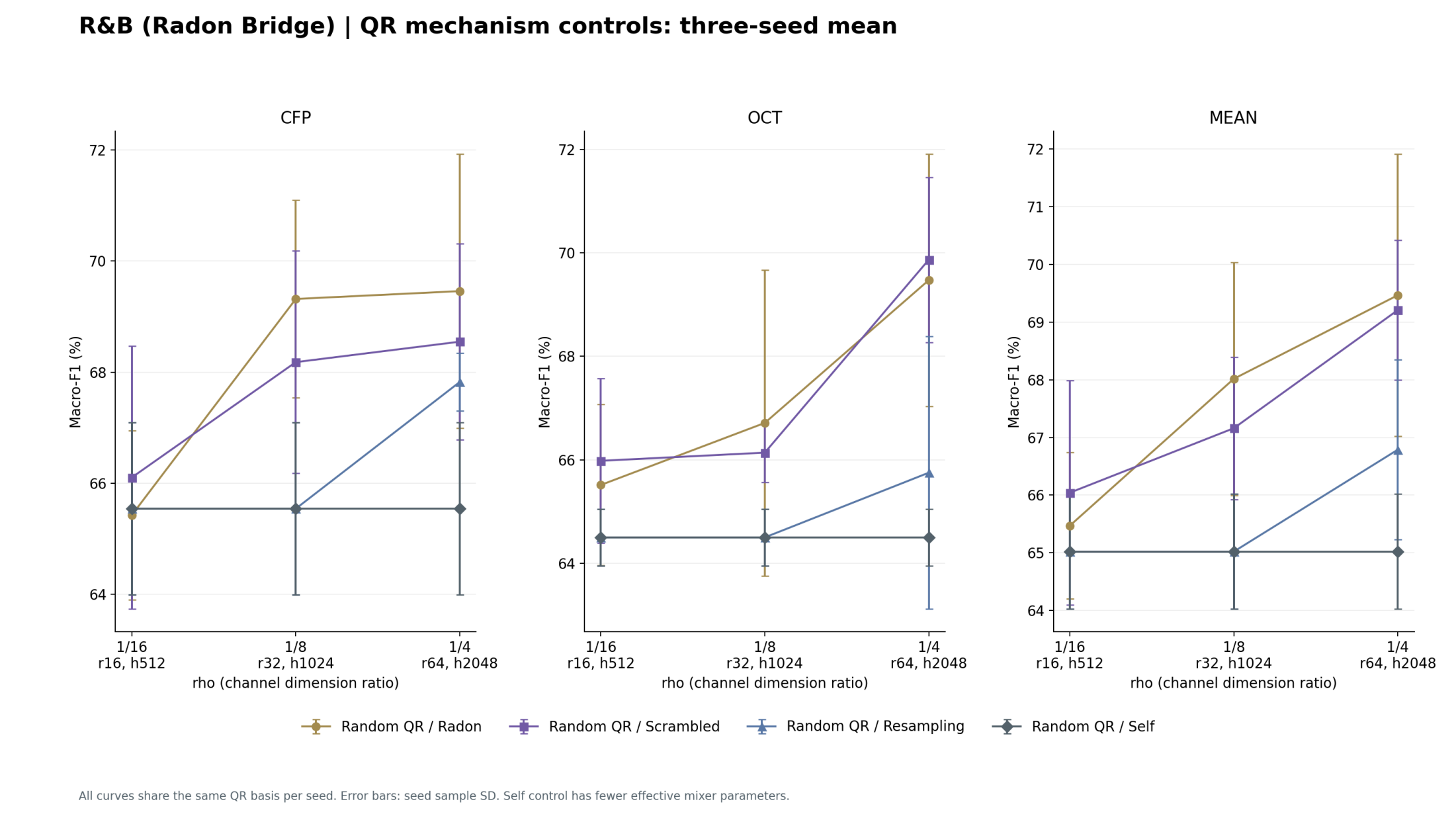
For SVD, the simultaneous interval[-7.17,-1.22] pp supports a substantive decline under the prespecified1 pp reference margin.

A zero spread across widths at the selected zero bridge is not evidence that the bridge learned a stable representation. Stability must be interpreted together with useful performance.

Do not select a later per-width checkpoint or change the training rule to rescue this result. The negative evidence is retained; the independent-width regime remains the working reference.

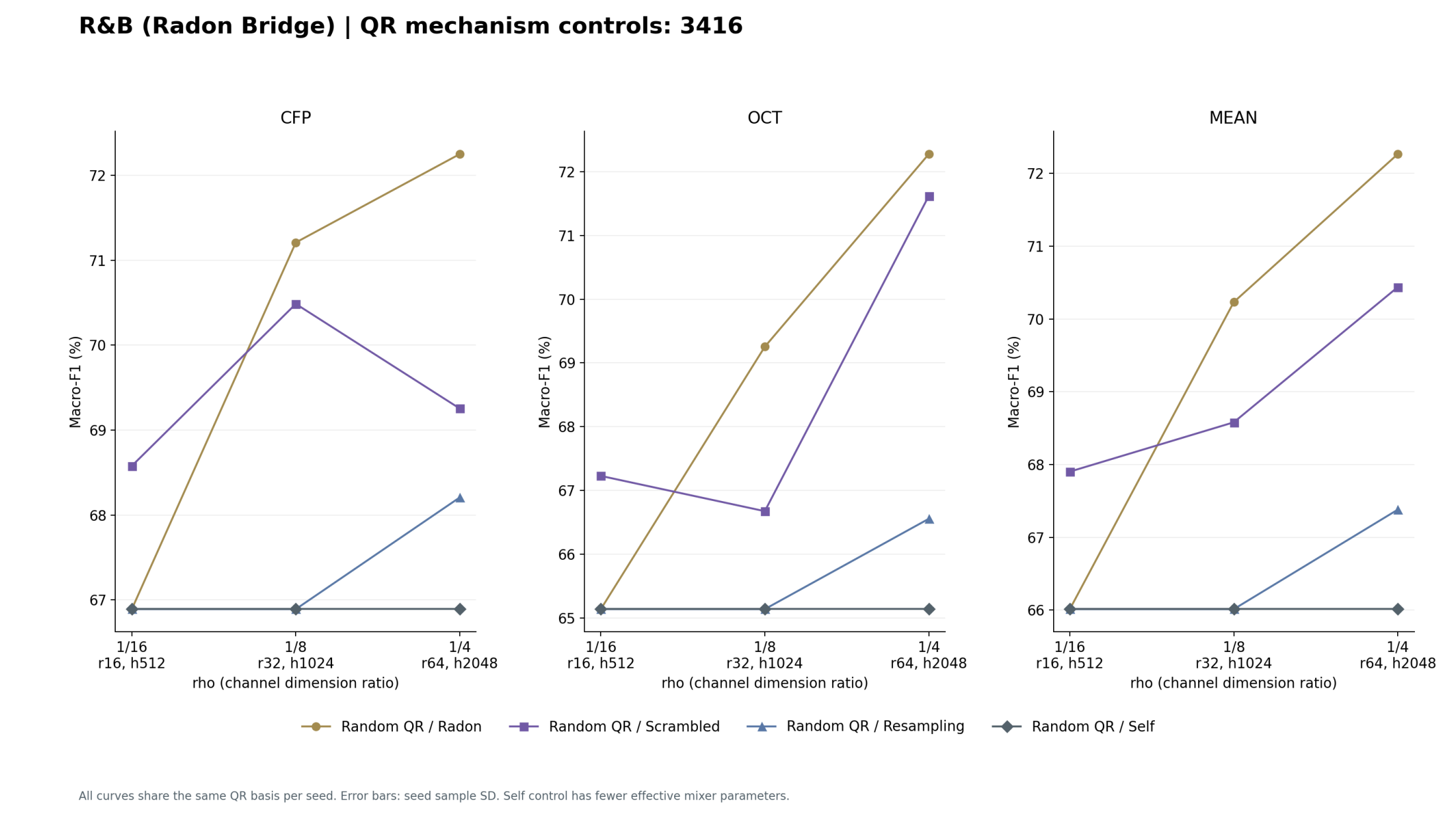
QR mechanism controls: three-seed mean

All curves share the same QR basis per seed. Error bars: seed sample SD. Self control has fewer effective mixer parameters.



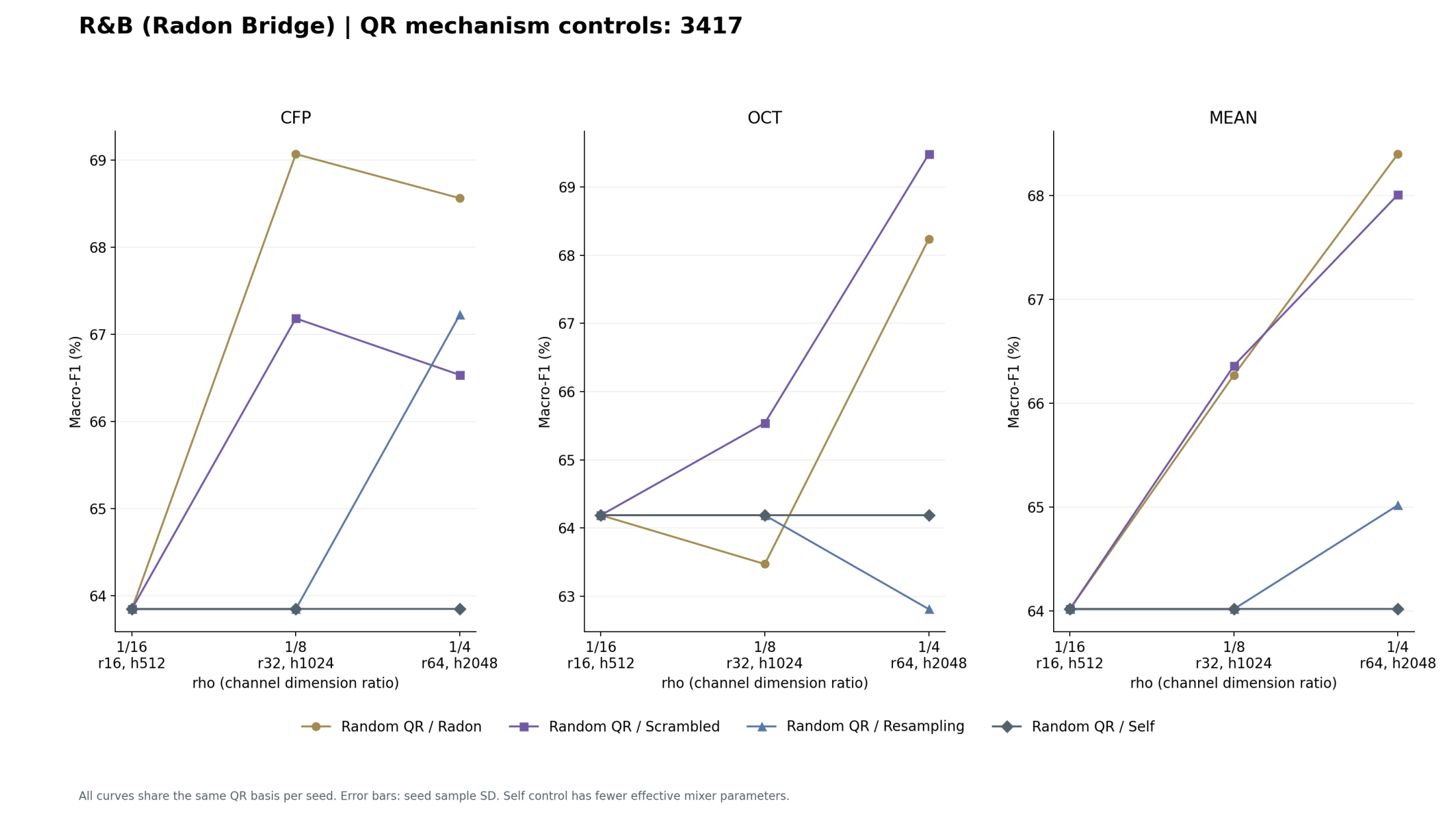
QR mechanism controls: 3416

All curves share the same QR basis per seed. Error bars: seed sample SD. Self control has fewer effective mixer parameters.



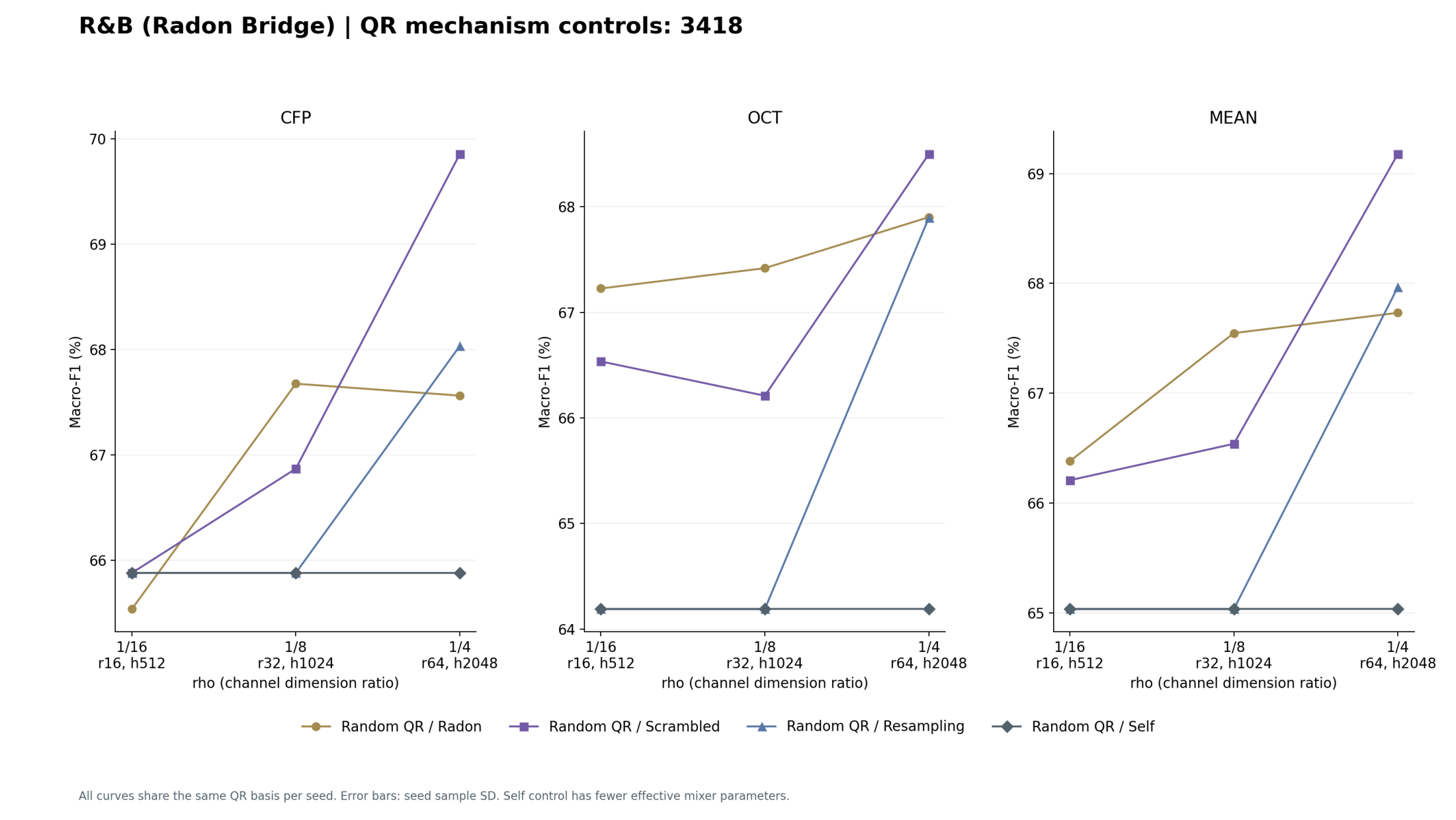
QR mechanism controls: 3417

All curves share the same QR basis per seed. Error bars: seed sample SD. Self control has fewer effective mixer parameters.



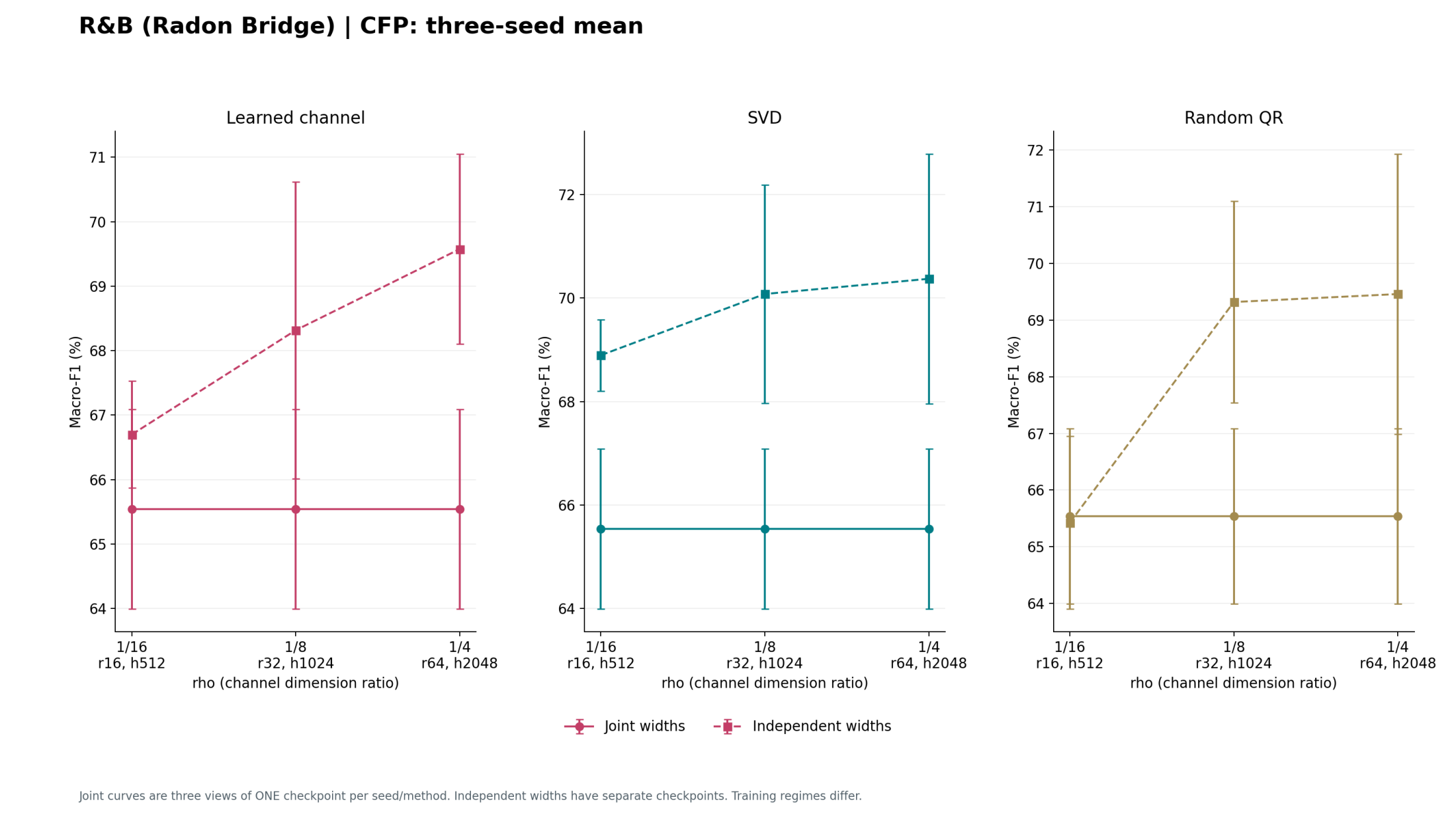
QR mechanism controls: 3418

All curves share the same QR basis per seed. Error bars: seed sample SD. Self control has fewer effective mixer parameters.



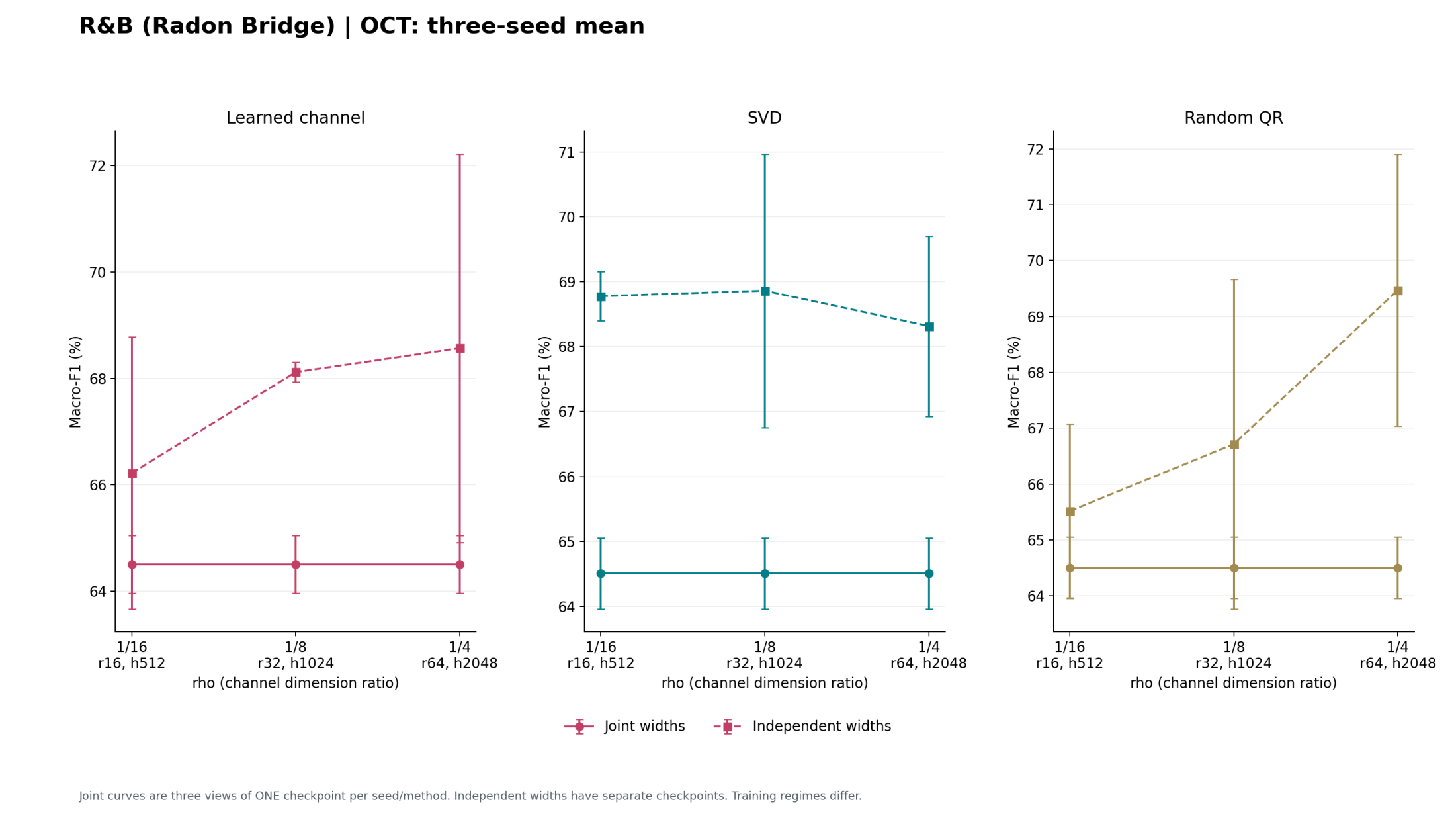
CFP: three-seed mean

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



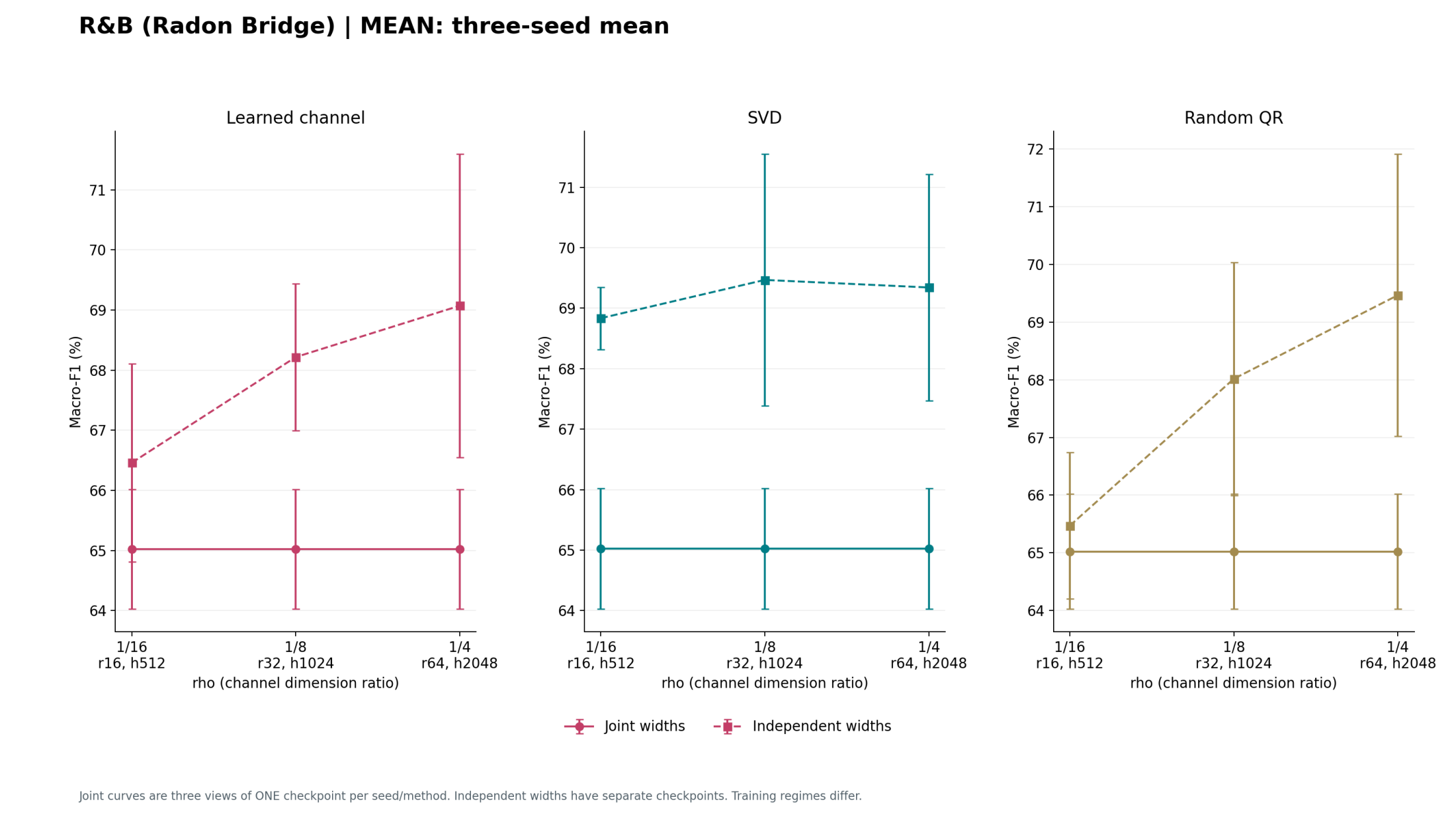
OCT: three-seed mean

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



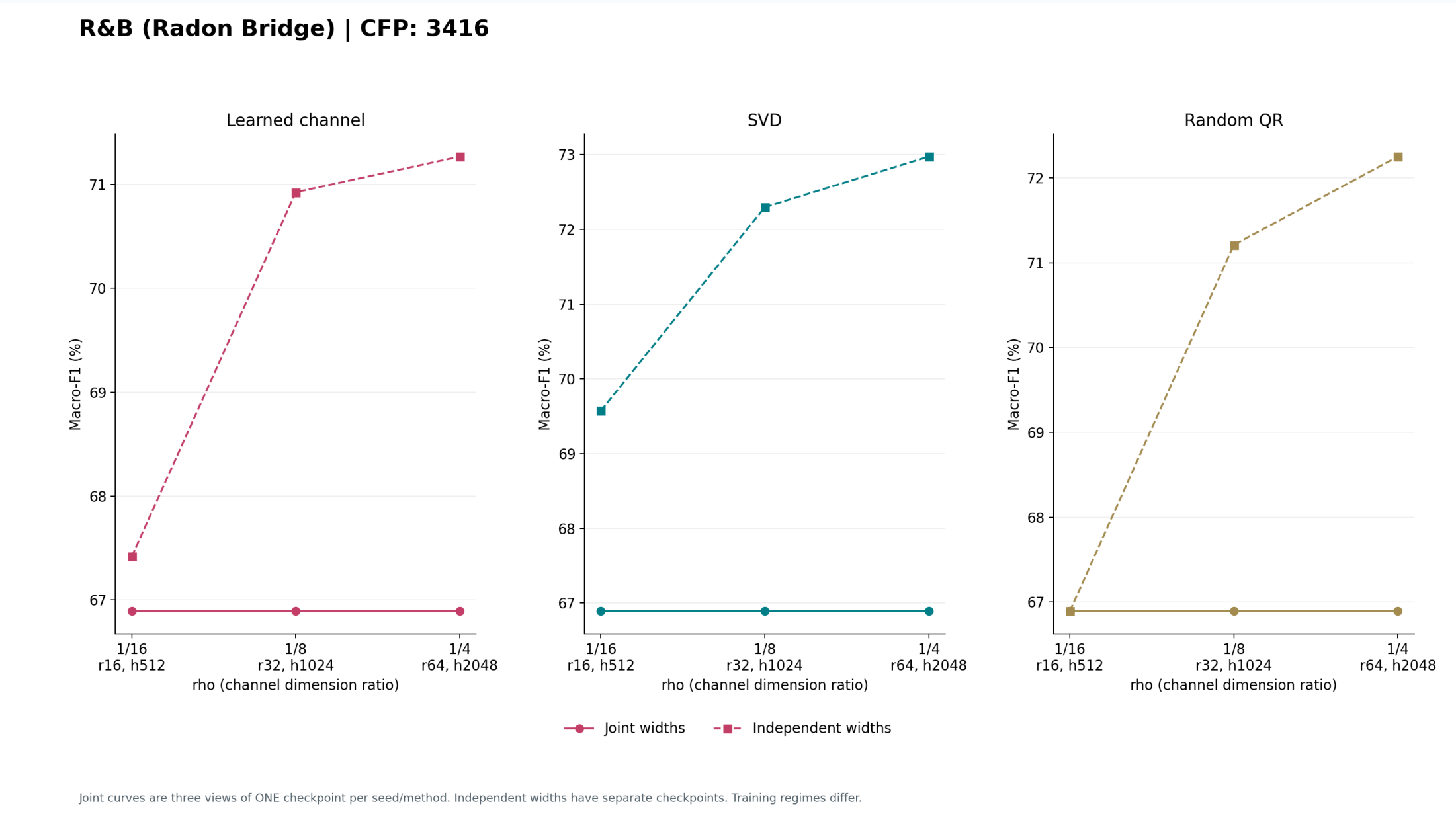
MEAN: three-seed mean

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



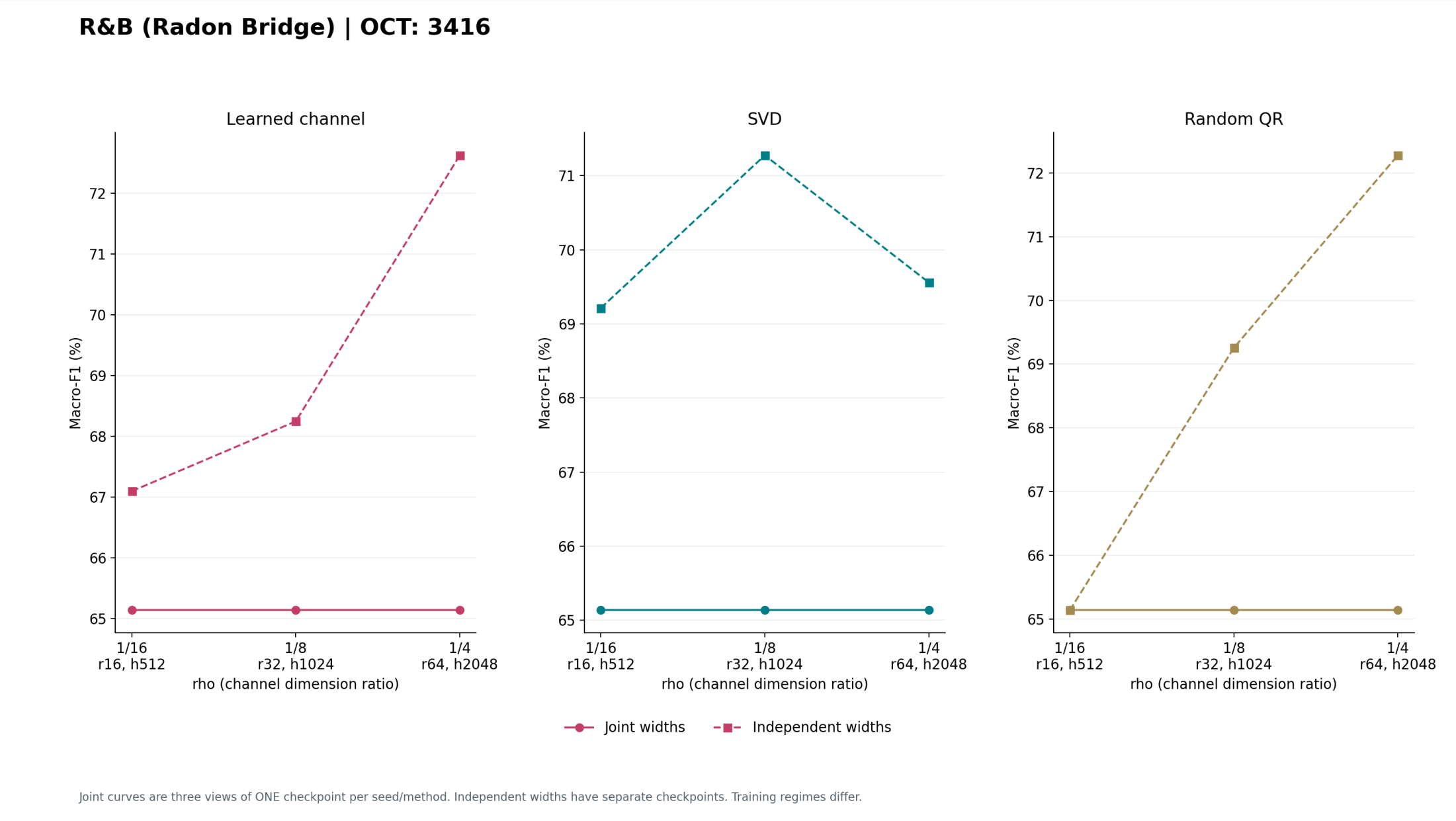
CFP: 3416

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



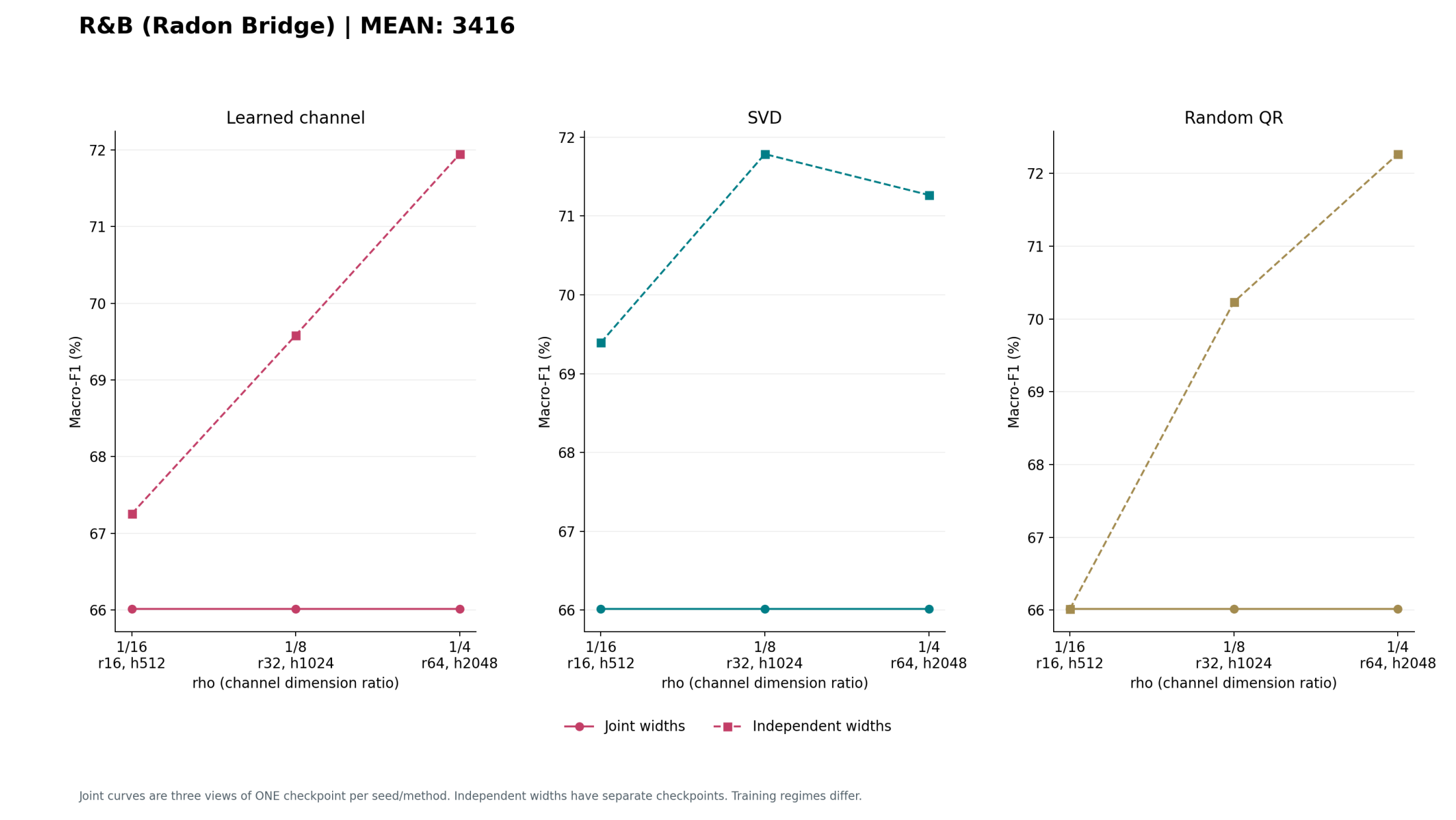
OCT: 3416

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



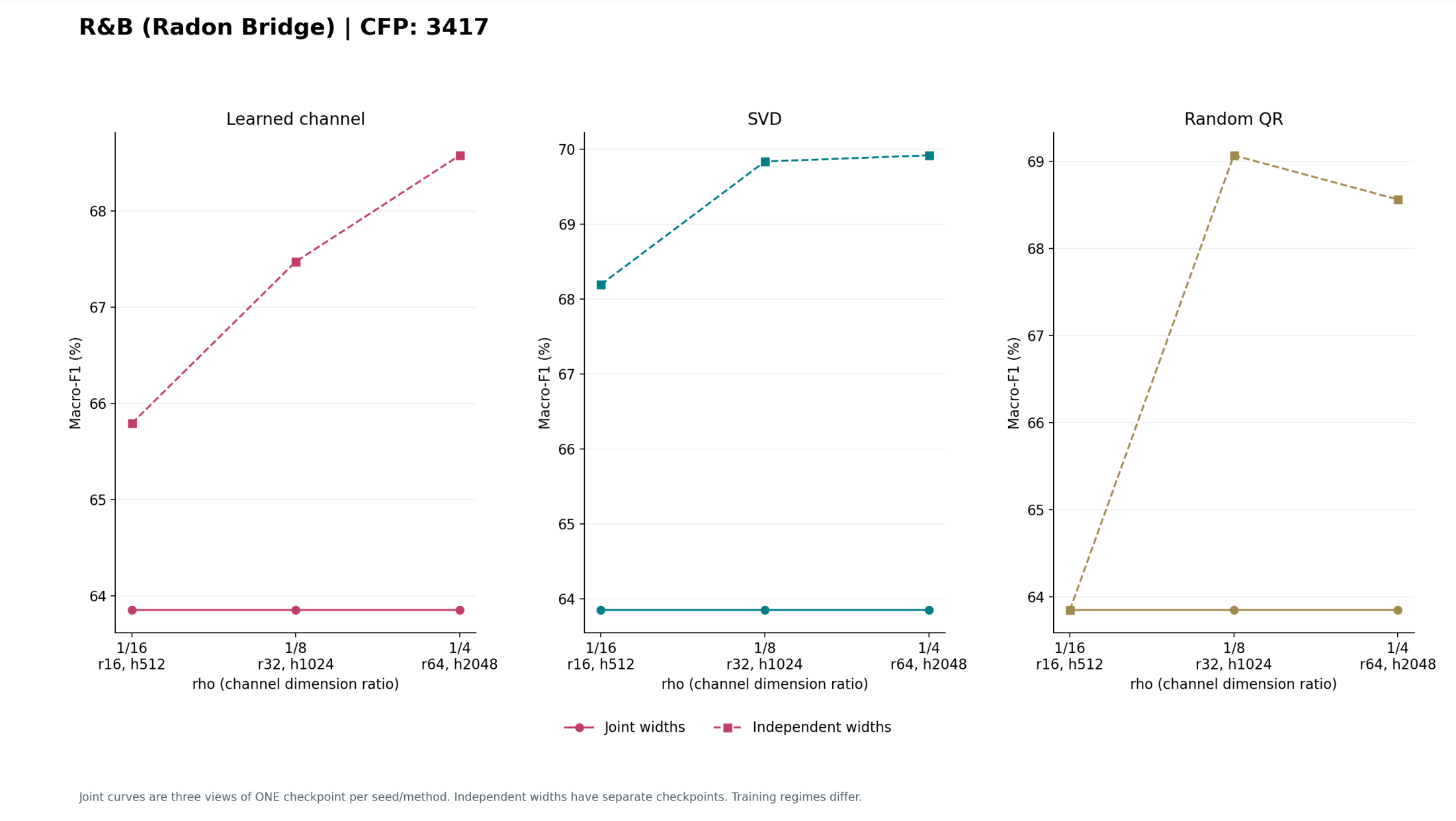
MEAN: 3416

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



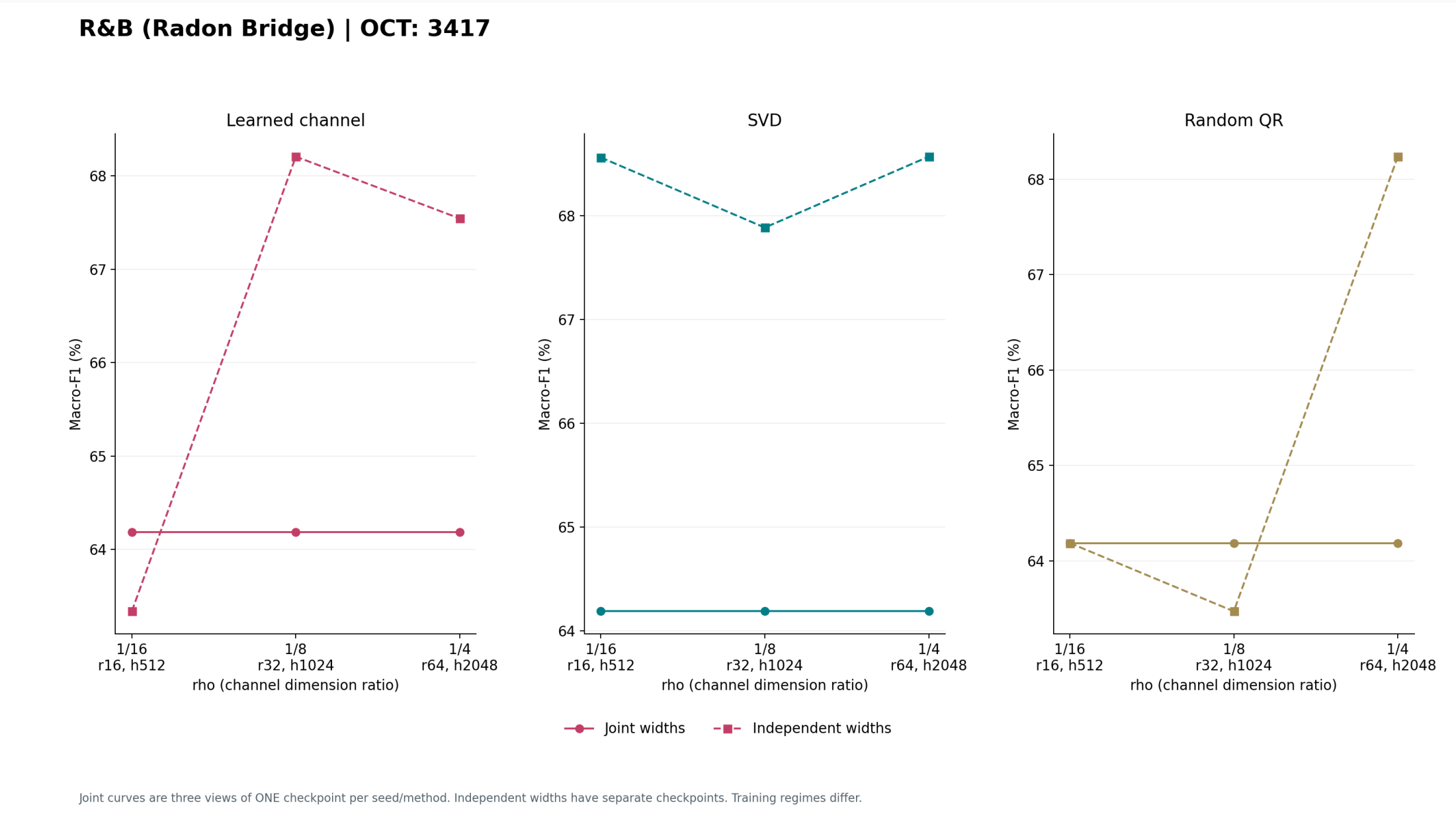
CFP: 3417

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



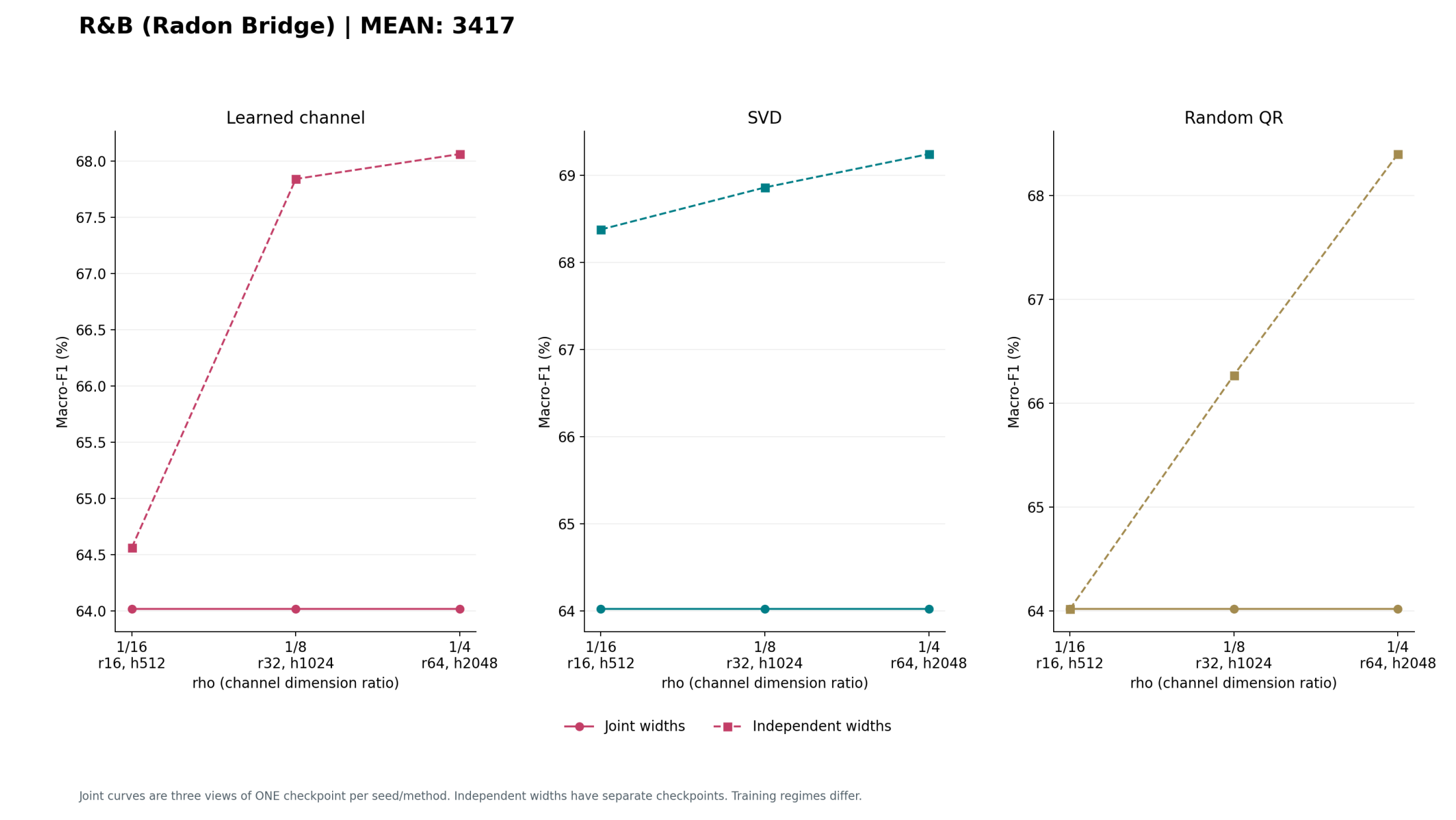
OCT: 3417

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



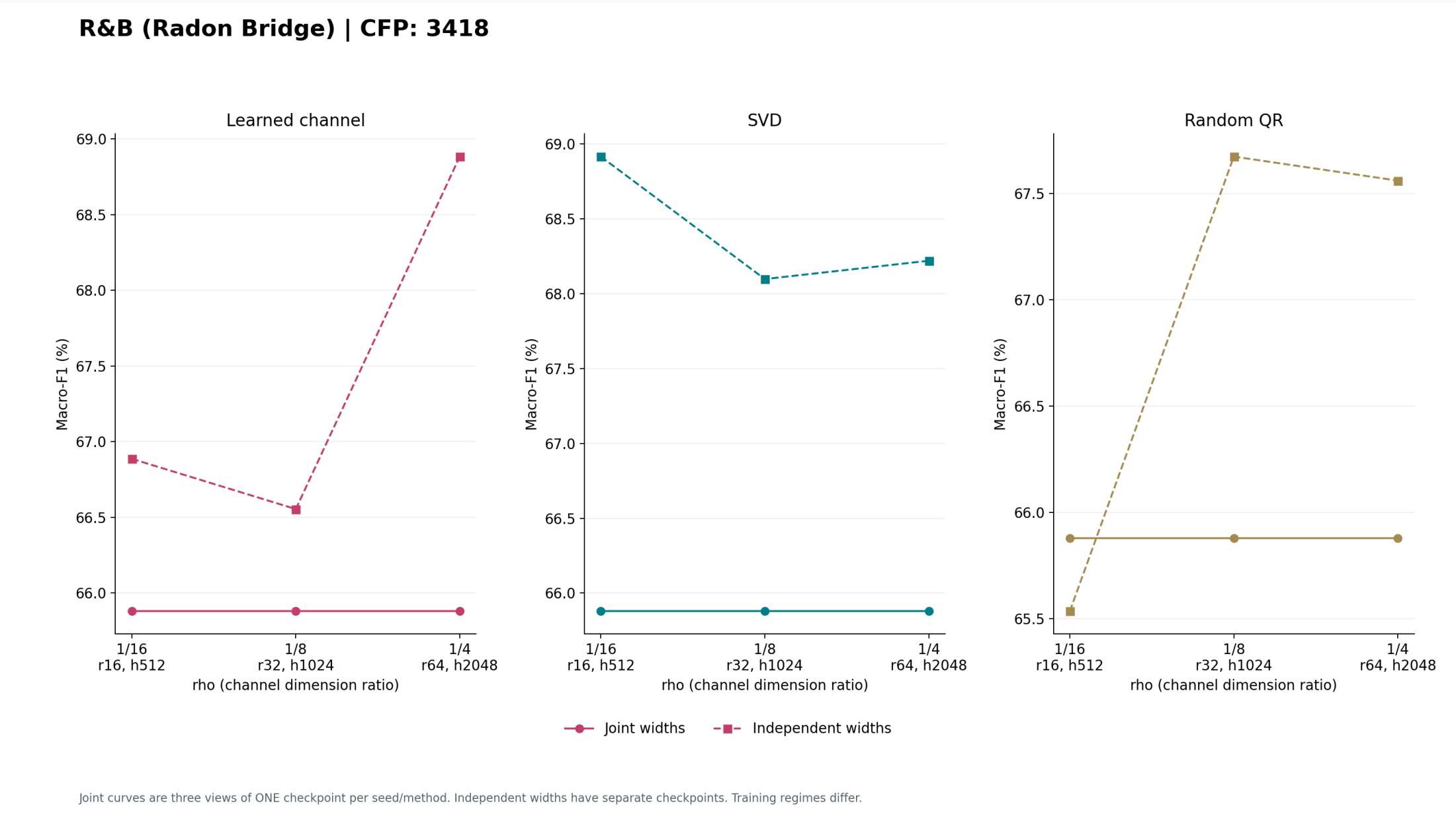
MEAN: 3417

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



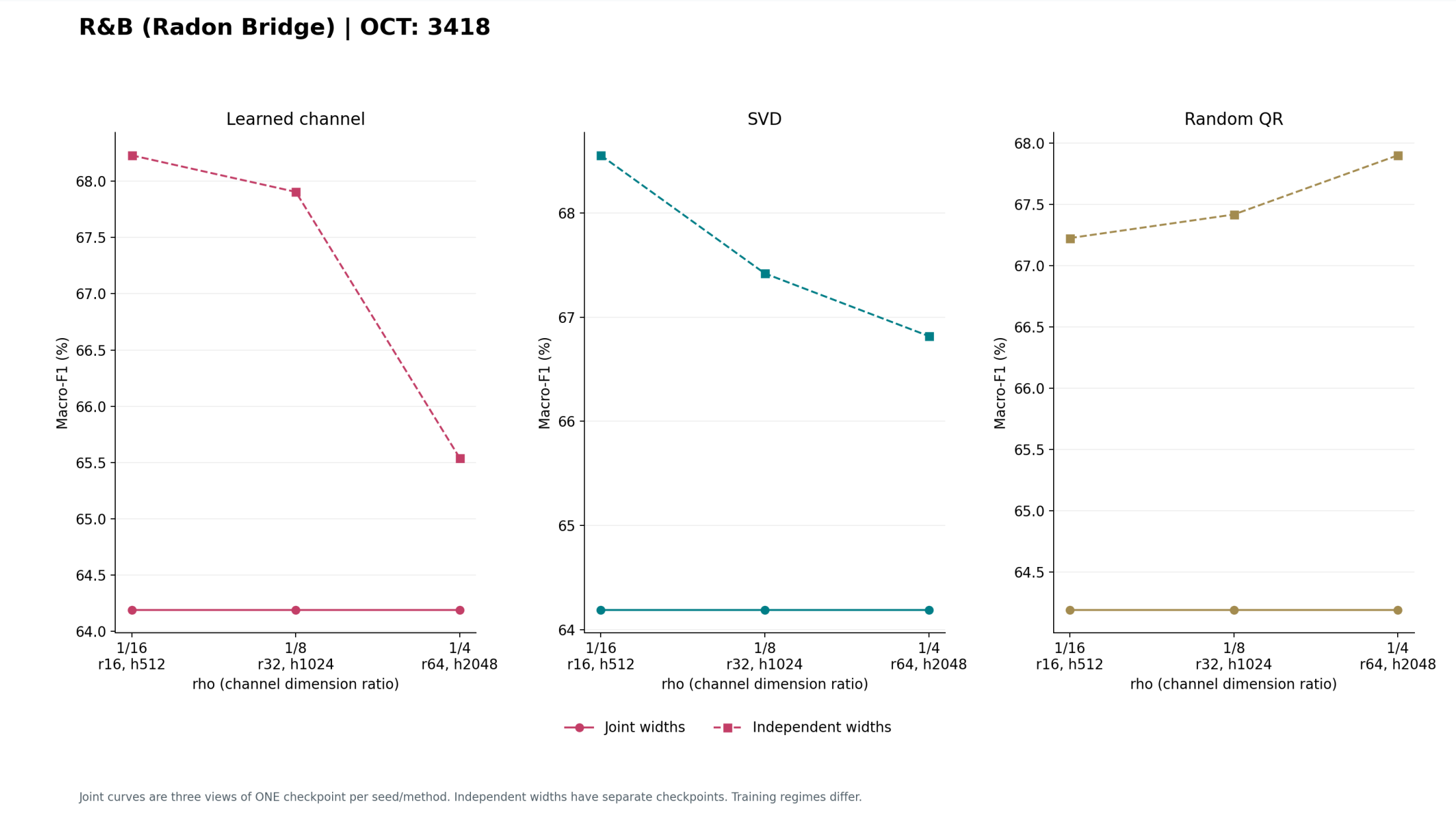
CFP: 3418

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



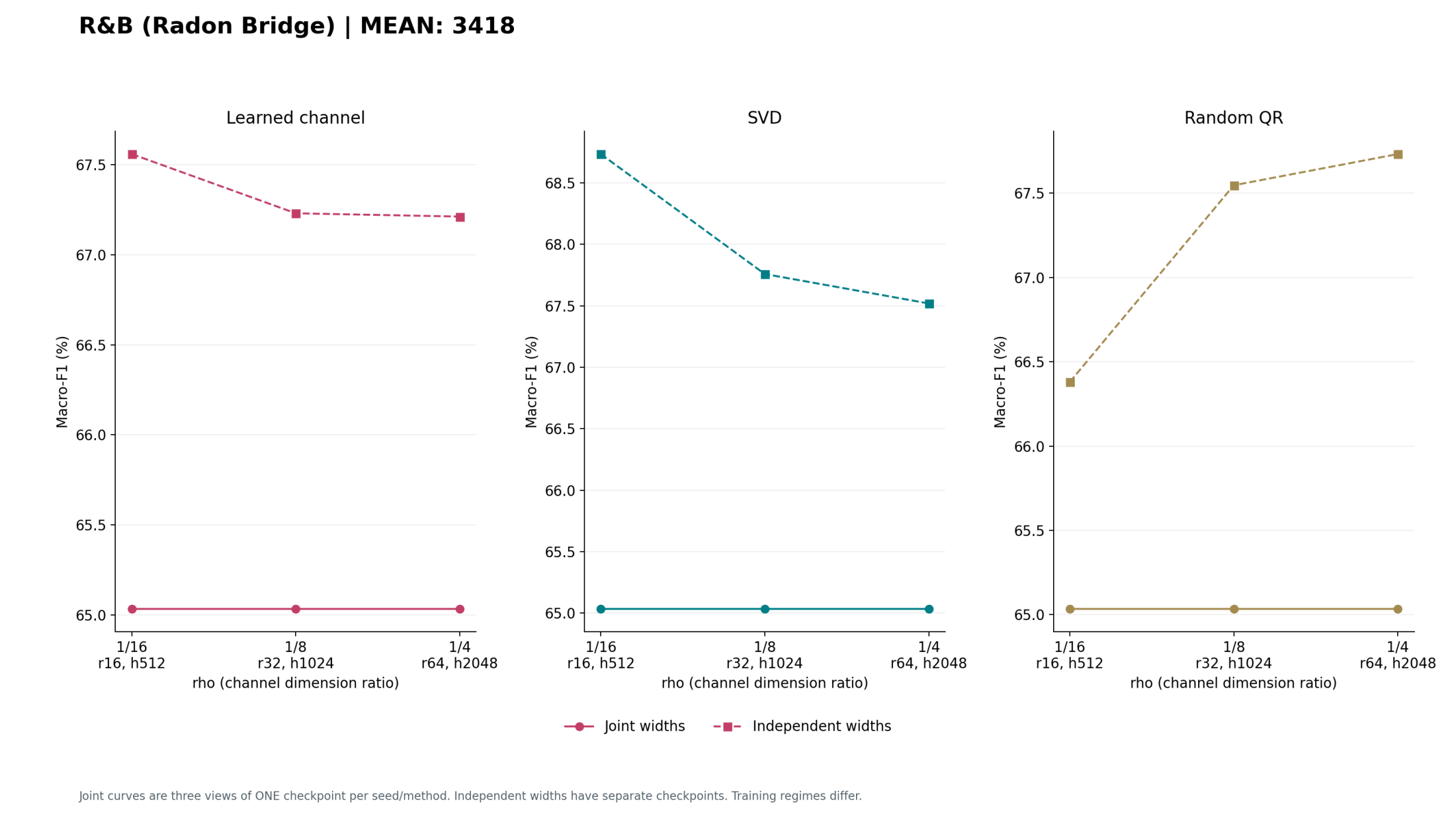
OCT: 3418

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



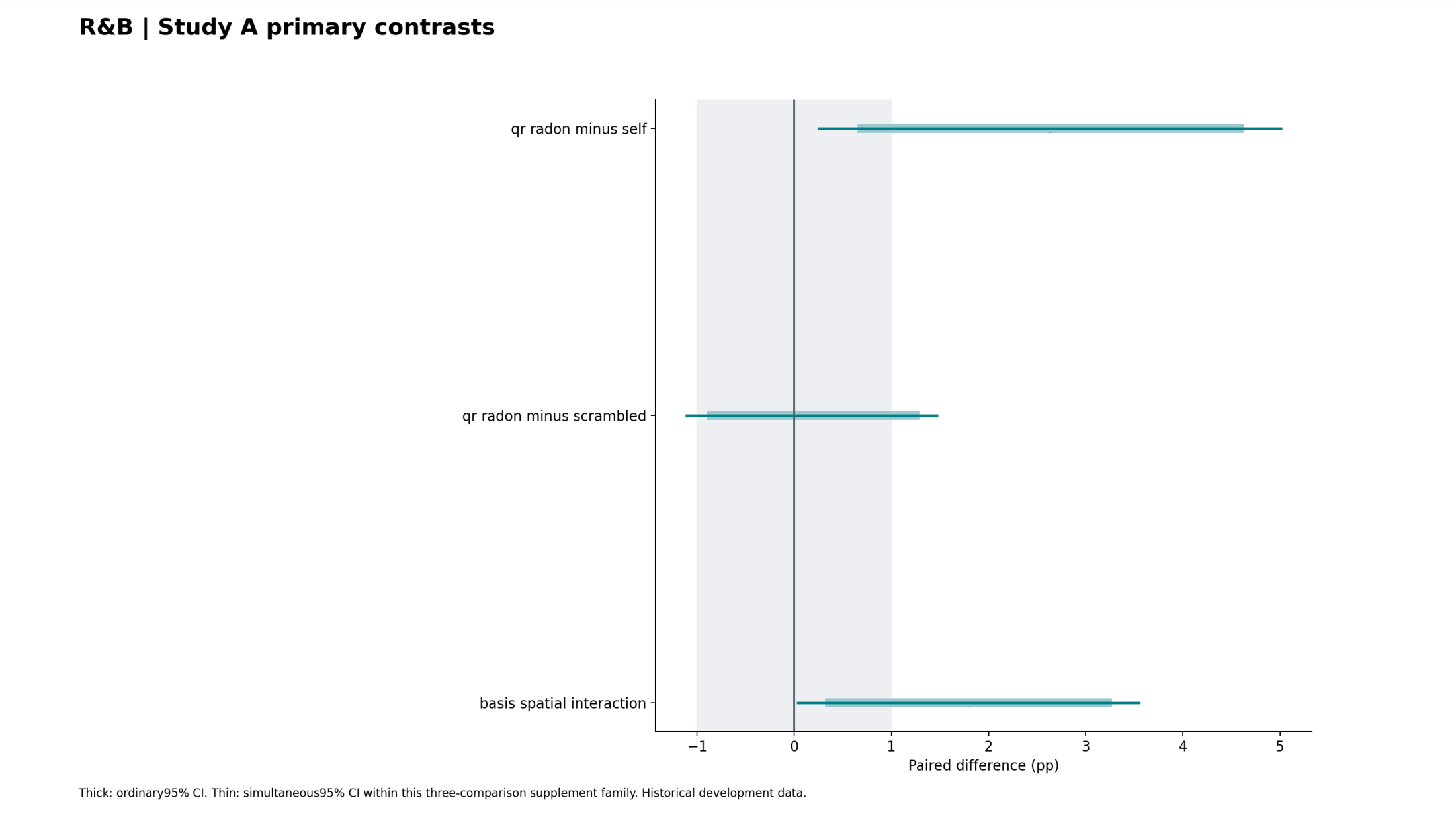
MEAN: 3418

Joint curves are three views of ONE checkpoint per seed/method. Independent widths have separate checkpoints. Training regimes differ.



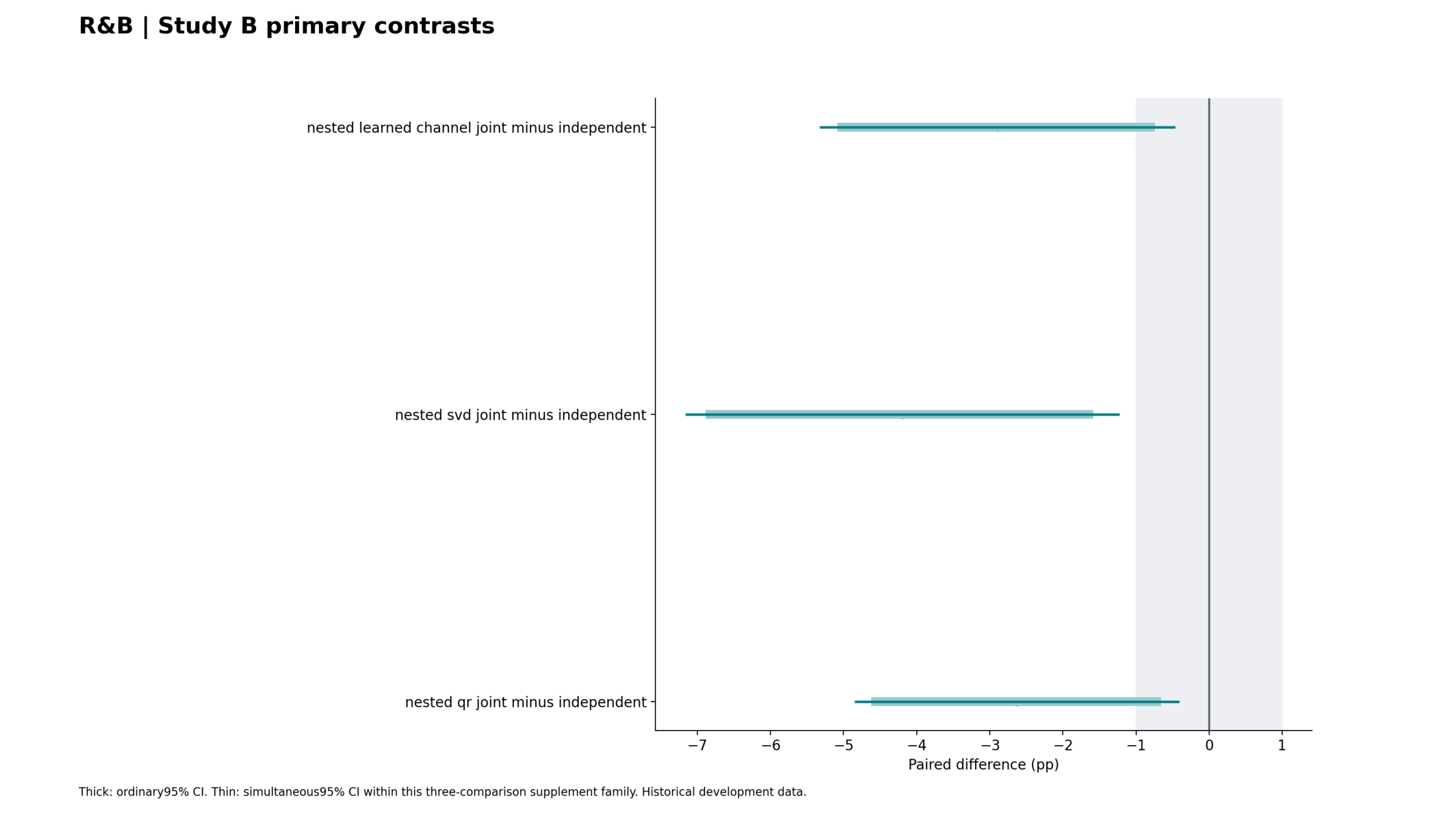
Study A primary contrasts

Thin: simultaneous interval; thick: ordinary95% interval. Reference band: +/-1 pp.



Study B primary contrasts

Thin: simultaneous interval; thick: ordinary95% interval. Reference band: +/-1 pp.



A: all18 new independent-width trainings

Rows1-9; F1 in percent. Best/stop epochs are preserved.

Seed	Arm / rho	CFP	OCT	Mean	Best / stop
3416	qr_self / 1:16	66.89	65.14	66.01	0 / 8
3416	qr_scrambled / 1:16	68.58	67.23	67.90	9 / 13
3416	qr_self / 1:8	66.89	65.14	66.01	0 / 8
3416	qr_scrambled / 1:8	70.49	66.67	68.58	10 / 16
3416	qr_self / 1:4	66.89	65.14	66.01	0 / 8
3416	qr_scrambled / 1:4	69.25	71.62	70.44	7 / 13
3417	qr_self / 1:16	63.85	64.19	64.02	0 / 8
3417	qr_scrambled / 1:16	63.85	64.19	64.02	0 / 8
3417	qr_self / 1:8	63.85	64.19	64.02	0 / 8

A: all18 new independent-width trainings

Rows10-18; F1 in percent. Best/stop epochs are preserved.

Seed	Arm / rho	CFP	OCT	Mean	Best / stop
3417	qr_scrambled / 1:8	67.18	65.54	66.36	13 / 19
3417	qr_self / 1:4	63.85	64.19	64.02	0 / 8
3417	qr_scrambled / 1:4	66.54	69.48	68.01	6 / 12
3418	qr_self / 1:16	65.88	64.19	65.03	0 / 8
3418	qr_scrambled / 1:16	65.88	66.54	66.21	17 / 17
3418	qr_self / 1:8	65.88	64.19	65.03	0 / 8
3418	qr_scrambled / 1:8	66.87	66.21	66.54	4 / 10
3418	qr_self / 1:4	65.88	64.19	65.03	0 / 8
3418	qr_scrambled / 1:4	69.86	68.50	69.18	17 / 23

B: all27 views of nine joint-width models

Rows1-9; F1 in percent. Best/stop epochs are preserved.

Seed	Arm / rho	CFP	OCT	Mean	Best / stop
3416	nested_learned_channel / 1:16	66.89	65.14	66.01	0 / 8
3416	nested_learned_channel / 1:8	66.89	65.14	66.01	0 / 8
3416	nested_learned_channel / 1:4	66.89	65.14	66.01	0 / 8
3416	nested_svd / 1:16	66.89	65.14	66.01	0 / 8
3416	nested_svd / 1:8	66.89	65.14	66.01	0 / 8
3416	nested_svd / 1:4	66.89	65.14	66.01	0 / 8
3416	nested_qr / 1:16	66.89	65.14	66.01	0 / 8
3416	nested_qr / 1:8	66.89	65.14	66.01	0 / 8
3416	nested_qr / 1:4	66.89	65.14	66.01	0 / 8

B: all27 views of nine joint-width models

Rows10-18; F1 in percent. Best/stop epochs are preserved.

Seed	Arm / rho	CFP	OCT	Mean	Best / stop
3417	nested_learned_channel / 1:16	63.85	64.19	64.02	0 / 8
3417	nested_learned_channel / 1:8	63.85	64.19	64.02	0 / 8
3417	nested_learned_channel / 1:4	63.85	64.19	64.02	0 / 8
3417	nested_svd / 1:16	63.85	64.19	64.02	0 / 8
3417	nested_svd / 1:8	63.85	64.19	64.02	0 / 8
3417	nested_svd / 1:4	63.85	64.19	64.02	0 / 8
3417	nested_qr / 1:16	63.85	64.19	64.02	0 / 8
3417	nested_qr / 1:8	63.85	64.19	64.02	0 / 8
3417	nested_qr / 1:4	63.85	64.19	64.02	0 / 8

B: all27 views of nine joint-width models

Rows19-27; F1 in percent. Best/stop epochs are preserved.

Seed	Arm / rho	CFP	OCT	Mean	Best / stop
3418	nested_learned_channel / 1:16	65.88	64.19	65.03	0 / 8
3418	nested_learned_channel / 1:8	65.88	64.19	65.03	0 / 8
3418	nested_learned_channel / 1:4	65.88	64.19	65.03	0 / 8
3418	nested_svd / 1:16	65.88	64.19	65.03	0 / 8
3418	nested_svd / 1:8	65.88	64.19	65.03	0 / 8
3418	nested_svd / 1:4	65.88	64.19	65.03	0 / 8
3418	nested_qr / 1:16	65.88	64.19	65.03	0 / 8
3418	nested_qr / 1:8	65.88	64.19	65.03	0 / 8
3418	nested_qr / 1:4	65.88	64.19	65.03	0 / 8

Audit trail, limitations and the next decision

Complete aggregate data and figure sources accompany this report; restricted records remain remote.

Accepted additions: 27 trainings and 45 initial/selected diagnostic views, with all training summaries converged by policy. 5 structures and 11 diagnostic views passed GPU preflight.

Total accounted GPU time 1560.62 minutes, including the preserved failed preflight attempt. Each project process remained subject to the original 10 GiB/card limit; batch 16 was unchanged.

Training source 6ada3a35dc7ea7a740e040c3cbd0623f7248218a. The failed preflight was repaired before production training without widening the numerical tolerance.

The overall independent-width ranking favors noncentered SVD on this development benchmark; configuration selection remains exploratory. No new method or parameter search is justified merely to obtain a positive result.

Next external questions require separately prespecified cohorts, tasks or models. Ethics, license, source exclusions and label timing remain author-verification items; this report does not assert clinical deployment validity.